


# 6MMCS009W1 Usability Testing & Evaluation

# Group Coursework


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## BSc (Hons) in Computer Science


School of Computer Science & Engineering

College of Design, Creative and Digital Industries
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# 1.Experimental Operandum

## 1.1Group information

| Student Name | Student Number |
|---|---|
| Kesavan Anbuchelvan | w1903041-20211452 |
| Naflan Niyas | w1899493 - 20211394 |
| Shanuka Karunarathne | w1870607 - 20210252 |
| Bimandee Premarathna | w1904454 - 20211467 |

*Table 1 Group Members Information (UT Group- 15)*

# 2. Introduction and Objectives

This experiment will be done to assess the usability of Google Workspace in terms of effectiveness, efficiency, and user satisfaction of all its features. As a group, we are supposed to find out what potential usability problems users encounter and how these problems affect their satisfaction. As such, the research objectives are to design and conduct a structured usability test to capture quantitative and qualitative data, analyse users' interactions with the platform and formulate evidence-based recommendations for improving the product's usability. The present study aims to shed light on the accessibility and usability of Google Workspace for different user groups.

# 3. Research Questions and Hypotheses

The research questions in the study were formed evaluating all five usability goals, Effectiveness, Efficiency, Engagement, Ease of Learning, and Error Tolerance (5Es). Each of the team members has developed one research question, in addition to the hypothesis that encompasses all four study techniques and five usability goals.

**Ho** = null hypothesis **Ha** = alternative hypothesis

**μ1** = mean of population 1 **μ2** = mean of population 2 **μ3**=mean of population 3

**ρ** = Pearson correlation of the population

## 3.1 Independent-sample-test

***Student Name***: **Bimandee Premarathna**

| | |
|---|---|
| **Research question** | How long does it typically take a student with prior knowledge to add an add-on compared to a student without previously acquired knowledge? |
| **Usability metric studied** | Efficiency |
| **Hypothesis (statement)** | **Null Hypothesis ($H_0$):** The amount of time that students with prior knowledge require on average to add an add-on is the same as that of students without prior knowledge. |
| | **Alternative Hypothesis ($H_1$):** The average time required to add an add-on differs based on prior knowledge. |

| | | |
|---|---|---|
| **Mathematical Expression** | **Null Hypothesis ($H_0$):** $\mu_1 = \mu_2$<br>$\mu_1$ = Mean time spent by students with prior knowledge<br>$\mu_2$ = Mean time spent by students without prior knowledge | |
| | **Alternative Hypothesis ($H_1$):** $\mu_1 \neq \mu_2$<br>$\mu_1$ = Mean time spent by students with prior knowledge<br>$\mu_2$ = Mean time spent by students without prior knowledge | |
| **Variables** | **Independent** | Prior knowledge about adding an add-on |
| | **Dependent** | Average time taken to add an add-on |
| | **Control** | Students focus solely on adding new add-ons to their workspace. |
| | **Measurement Scale** | Time measured in seconds (ratio scale) |
| | **Data type** | Quantitative: Time taken to complete the task is measured numerically (in seconds), making it a continuous quantitative variable. |
| **Experimental design** | **Between-Subjects Design**: Each participant belongs to only one group (either with or without prior knowledge). This design avoids potential carryover effects. | |

| | |
|---|---|
| **Tasks and Scenarios** | Participants will complete the task:<br><br>Adding the "Form Builder for Google Sheets" add-on to the sheet with the data provided, and making a google form out of it. |
| **Procedure** | Data will be collected **in person**, in a controlled lab environment to minimize external distractions. The sessions will be **unmoderated**, with no need for interference, allowing participants to complete the task independently while still following the provided instructions. |
| **Data collection method** | Tools: Participants used a timer to log their time during the task. This tool is accurate and reliable for measuring the primary metric (time). |
| **Test material** | Materials:<br><br>1. Task instructions for participants<br>2. Questionnaire to gather demographic information and prior knowledge<br><br>Measurement:<br><br>1. Task start: when the participant begins interaction.<br>2. Task end: when the add-on is successfully added, google form is formed. |

*Table 2 Independent-sample-test Research Question and Hypothesis table*

## 3.2 Paired-sample-test

***Student Name:*** **Naflan Niyas**

<table>
<tr><td>Research question</td><td colspan="3">What effects does the "Form Builder Plus" add-on have on students' workflow when they switch from Google Docs to Google Forms?</td></tr>
<tr><td>Usability metric studied</td><td colspan="3">Engagement</td></tr>
<tr><td rowspan="2">Hypothesis (statement)</td><td colspan="3">Null Hypothesis ($H_0$): Students' workflow involvement is not significantly impacted by switching from Google Docs to Google Forms with installed third-party add-ons. (Form Builder Plus)</td></tr>
<tr><td colspan="3">Alternative Hypothesis ($H_1$): Students' workflow involvement is significantly impacted by this transition.</td></tr>
<tr><td rowspan="2">Mathematical Expression</td><td colspan="2">Null Hypothesis ($H_0$): $\mu_1 = \mu_2$</td><td rowspan="2">$\mu_1$: Mean engagement level during the transition using the add-on.<br><br>$\mu_2$: Mean engagement level without the transition or add-on.</td></tr>
<tr><td colspan="2">Alternative Hypothesis ($H_1$): $\mu_1 \neq \mu_2$</td></tr>
<tr><td rowspan="3">Variables</td><td>Independent</td><td colspan="2">Transition from Google Docs to Google Forms using a third-party add-on. (Form Builder Plus)</td></tr>
<tr><td>Dependent</td><td colspan="2">Students' workflow engagement.</td></tr>
<tr><td>Control</td><td colspan="2">Participants focus on one specific third-party add-on ("Form Builder Plus").</td></tr>
</table>

| | | |
|---|---|---|
| | **Measurement Scale** | Use a **Likert scale** (ordinal scale) to measure perceived engagement (e.g., 1–5 for "low to high engagement") or a **ratio scale** for task completion time and the number of interactions recorded. |
| | **Data type** | **Quantitative Data:** Engagement metrics such as time spent on tasks or Ordinal Scale ratings are numerical. Time is ratio data (continuous), and Ordinal Scale ratings are ordinal data, both quantifiable. |
| **Experimental design** | **Within-Subjects Design** (Paired-Sample): Each participant completes tasks in two conditions (with and without the add-on). This ensures any differences in engagement are due to the add-on rather than individual variability. | |
| **Tasks and Scenarios** | Participants will complete two tasks:<br>1. Build a form using Google Docs and transition to Google Forms **with the add-on**.<br>2. Build the same form without using the add-on. Engagement levels will be measured for both | |
| **Procedure** | Data will be collected **in person** in a lab environment to control for distractions. The sessions will be **moderated** to provide participants with consistent instructions and ensure proper use of the add-on. | |
| **Data collection method** | **Tools:** Post-test surveys are tools used to accurately capture quantitative data on task completion time and user perceptions of engagement. | |
| **Test material** | **Materials:** | |

| | |
|---|---|
| | 1. Instructions for tasks provided in document and verbal formats.<br>2. Task environment (Google Docs and Google Forms).<br><br>**Measurement:** Task starts when participants begin interacting with the interface and ends upon successful task completion. |

*Table 3 Paired Sample Test Research Question and Hypothesis table*

## 3.3 ANOVA

***Student Name***: **Kesavan Anbucvhelvan**

| | | |
|---|---|---|
| **Research question** | Do the sidebar add-ons' preventative strategies **("About," "Feedback," "Help"**) differ significantly in their ability to assist students in minimizing errors when using particular add-ons? | |
| **Usability metric studied** | Error Tolerance | |
| **Hypothesis (statement)** | **Null Hypothesis ($H_0$):** There is no significant difference in the helpfulness of the prevention methods (About, Help, Feedback) provided by the sidebar add-ons for students to avoid errors while using specific add-ons.<br><br>**Alternative Hypothesis ($H_1$):** There is a significant difference in the helpfulness of the prevention methods (About, Help, Feedback) provided by the sidebar add-ons for students to avoid errors while using specific add-ons. | |
| **Mathematical Expression** | **Null Hypothesis Ho: μ1 = μ2 = μ3** | **$\mu_1$:** Measure of usefulness for "About" |

| | | | |
|---|---|---|---|
| | | | **$\mu_2$:** Measure of usefulness for "Help"<br><br>**$\mu_3$:** Measure of usefulness for "Feedback" |
| | **Alternative Hypothesis Ha:**<br><br>**Ha: μ1 ≠ μ2 or**<br><br>**Ha: μ1 ≠ μ3 or**<br><br>**Ha: μ2 ≠ μ3** | | **$\mu_1$:** Measure of usefulness for "About"<br><br>**$\mu_2$:** Measure of usefulness for "Help"<br><br>**$\mu_3$:** Measure of usefulness for "Feedback" |
| **Variables** | **Independent** | Preventative techniques ("Help," "Feedback," "About") | |
| | **Dependent** | Level of assistance provided (measured through error minimization or error count). | |
| | **Control** | Students are only exposed to these three predefined options. | |
| | **Measurement Scale** | Use an **Ordinal scale** (e.g., 1–5) to assess perceived usefulness or **error count** (ratio scale) to measure actual error minimization. | |
| | **Data type** | **Quantitative**: The effectiveness of each technique is measured either by error count (continuous) or Ordinal scale ratings (ordinal). | |

| | | |
|---|---|---|
| | | These are quantitative as they involve numerical representations of data. |
| **Experimental design** | **Within-Subjects Design:** Each participant evaluates all three options (About, Feedback, Help) to reduce variability caused by individual differences. This design ensures that differences are attributable to the options rather than the participants. | |
| **Tasks and Scenarios** | Participants will perform a task involving the use of an add-on (e.g., completing a form or configuring a setting). Each task will require them to use one of the three options ("About," "Help," or "Feedback") for assistance. | |
| **Procedure** | Data was collected **in person** at the IIT Computing School's computer lab. Tasks were predefined, and participants completed them using their local setups. The sessions were **unmoderated**, ensuring that researcher influence was minimized while participants followed clearly outlined instructions. provided in a digital format at the start of the session. | |
| **Data collection method** | **Tools:** Error self-reporting was implemented through post-test surveys, where participants were asked to reflect on any errors encountered during the tasks. These surveys collected quantitative data on user ratings for perceived effectiveness, and engagement with the sidebar options. Additionally, a stopwatch or timer software was used to measure the time taken to complete each task, providing a quantitative metric to further assess usability if needed. | |
| **Test material** | **Materials:**<br>1. Task instructions provided in a structured format. | |

| | |
|---|---|
| | 2. A post-task questionnaire to gather participant ratings for each option.<br><br>**Measurement:**<br>1. Task starts when the participant opens the sidebar option.<br>2. Task ends when the add-on task is completed successfully. |

*Table 4 ANOVA Testing Research Question and Hypothesis table*

## 3.4 Pearson Correlation

***Student Name:*** **Shanuka Karunarathne**

| | |
|---|---|
| **Research question** | Does the amount of time students spend studying the add-ons and the amount of time they spend working together on Google Docs assignments correlate? |
| **Usability metric studied** | Ease of learning |
| **Hypothesis (statement)** | **Null Hypothesis ($H_0$):** The amount of time students spends studying the add-ons and the amount of time they spend working together on Google Docs assignments do not significantly correlate ($\rho = 0$).<br><br>**Alternative Hypothesis (Ha):** The amount of time students spends studying the add-ons and their time spent working together on Google Docs assignments are significantly correlated ($\rho \neq 0$). |

| | | | |
|---|---|---|---|
| **Mathematical Expression** | **Null Hypothesis**<br><br>**($H_0$): ρ = 0** (no correlation) | | Where ρ = Pearson correlation coefficient. |
| | **Alternative Hypothesis (Ha): ρ ≠ 0** | | Where ρ = Pearson correlation coefficient. |
| **Variables** | **Independent** | Time spent studying the add-ons. | |
| | **Depended** | Time spent working together on assignments. | |
| | **Control** | Students will complete specific tasks after learning about the add-ons. | |
| | **Measurement Scale** | Use an **Ordinal scale** for both variables (e.g., time measured in seconds or minutes). Ratio scales are ideal because they have a true zero point and equal intervals. | |
| | **Data type** | Time spent on studying add-ons and working on assignments is measurable in numerical terms (e.g., minutes), making it continuous and quantitative. | |
| **Experimental design** | **Within-Subjects Design**<br><br>Each participant goes through both tasks:<br><br>1. **Learning the Add-On**: Time spent studying the add-on.<br>2. **Working Collaboratively**: Time spent completing the assignment in Google Docs. | | |

| | |
|---|---|
| | The same group of participants performs both tasks, allowing you to measure correlations between the two variables directly for each individual. |
| **Tasks and Scenarios** | Participants will:<br>1. Spend time learning how to use a specific add-on.<br>2. Work collaboratively on a predefined assignment in Google Docs that requires the add-on. Their time for both activities will be recorded. |
| **Procedure** | Data collection will be done **in person** using screen recording tools to capture timestamps. This will be **unmoderated** to reduce the influence of observers on participants' natural workflow. |
| **Data collection method** | **Tools:** Participants used a timer to log their time during the task. This tool is accurate and reliable for measuring the primary metric (time). |
| **Test material** | **Materials:**<br>1. A tutorial for the add-on (to measure time spent learning it).<br>2. An assignment in Google Docs (to measure time spent completing it collaboratively).<br>**Measurement:** Task completion is measured from start to end for each activity. |

*Table 5 Pearson Correlation Testing Research Question and Hypothesis table*

# 4. Study planning and organisation

This study was structured to explore the formulated research questions and hypotheses using a mix of **moderated** and **unmoderated** usability tests, all conducted in a physical setting. The primary focus was on assessing the usability of add-ons available in Google Workspace applications. Participants were students from the **Informatics Institute of Technology (IIT)**, affiliated with the University of Westminster and enrolled across various academic levels.

We chose **Google Workspace add-ons** as the focus of this study due to their widespread usage and integration into academic and professional workflows. Google Workspace applications like Docs, Forms, and Sheets are commonly used for tasks such as document creation, form creation, collaboration, and data management. The add-ons extend the core functionalities of these applications, enabling users to perform advanced operations such as automating tasks, customizing templates, and enhancing productivity. For students, these add-ons can streamline academic tasks, foster collaboration, and improve efficiency in managing group projects or assignments. By evaluating the usability of the sidebar options—such as "About," "Help," and "Feedback" we aimed to understand how effectively these features assist users in minimizing errors and navigating the add-ons. This focus aligns with the practical needs of our participant group and underscores the importance of intuitive and error-tolerant interfaces in commonly used tools.

To address the research questions effectively, careful planning was essential. Moderated tests were organized for specific questions requiring direct observation and guidance. These sessions took place at the **IIT Computing School building**, where participants received a detailed briefing about the tasks, relevant resources (e.g., application links), and clear instructions. Moderators actively supported participants during these sessions, ensuring tasks were completed accurately and without bias. After the tasks, participants responded to post-task questionnaires distributed via Google Forms.

On the other hand, unmoderated tests involved sharing questionnaires through email and text messages. Participants completed these tasks independently over a span of three days, with their responses collected remotely. To eliminate potential biases, all

participants were provided with an information sheet and a consent form, ensuring they understood the process and their role in the study.

Students from IIT were selected as participants because they represent the primary user base for Google Workspace applications and their add-ons. Their frequent use of these tools for academic purposes, such as completing assignments, collaborating on projects, and managing academic workflows, makes them a highly relevant group for evaluating the usability and effectiveness of sidebar features. Additionally, their varied academic levels offer diverse perspectives on usability, ranging from novice to more experienced users.

A total of **20 students** participated in the study, divided as follows:

| Level Of Study | Number of Students |
|---|---|
| Level4 | 05 |
| Level5 | 05 |
| Placement students | 05 |
| Level6 | 05 |
| Total Participant Count | 20 |

***Table 6 Participant Count Table***

Each group member contributed to different aspects of the study, including participant recruitment, providing task briefings, moderating testing sessions, and collecting data. Every member ensured that their assigned research questions were appropriately addressed, aligning their efforts with the overall study framework.

The study spanned one week, with the testing phase completed over three days. Google Forms was used to create and distribute questionnaires for both moderated and unmoderated tests. The collaboration and active involvement of all group members played a crucial role in the study's success, ensuring all activities were carried out effectively and on schedule

## 4.1 Experimental Design

### 4.1.1 Experimental Design for Research Question 1 (Bimandee)

This research question used a **between-subjects design** to evaluate the **influence of prior knowledge** on the **time taken** to add an add-on in Google Sheets. This design was selected to avoid potential carryover effects and isolate the impact of prior knowledge on task performance.

The independent variable was prior knowledge (students with vs. without prior knowledge), and the dependent variable was the time taken to complete the task, measured from the initiation of the installation process to its successful completion. Participants were assigned one main task: to install the "Form Builder for Sheets" add-on and use it to create a Google Form from a dataset.

The tasks were **unmoderated** and conducted in a controlled lab environment to minimize distractions. Participants followed a standardized instruction sheet and recorded their start and end times. Data was collected using timers and post-task questionnaires to capture completion times and participant feedback.

By dividing participants into distinct groups and maintaining consistent task conditions, the between-subjects design effectively enabled a direct comparison of **task efficiency** based on prior knowledge.

### 4.1.2 Experimental Design for Research Question 2 (Naflan)

A **within-subjects design** was used to assess how the "Form Builder Plus" add-on impacts students' workflow during the transition from Google Docs to Google Forms. Each participant performed tasks under two conditions: with and without the add-on. This paired-sample approach was chosen to minimize individual differences and focus on the add-on's effects.

The independent variable was the **use of the add-on**, and the dependent variable was **workflow engagement**, measured through metrics such as task completion time, interaction logs, and Likert scale ratings. Participants completed two tasks—one with the add-on and one without—and their engagement levels were compared.

The experiment was conducted in a lab environment to control for distractions. Moderators ensured participants understood the tasks and provided consistent instructions. Data was collected using engagement tracking tools and post-test surveys, capturing both quantitative and qualitative insights. This design allowed for a direct comparison of workflow engagement with and without the add-on, aligning with the research question's objectives.

### 4.1.3 Experimental Design for Research Question 3 (Kesavan)

This research question used a **within-subjects design** to evaluate the effectiveness of three sidebar options **"About," "Feedback,"** and **"Help"** in minimizing errors. This design was selected to control for variability among participants, ensuring that observed differences were due to the sidebar options rather than individual characteristics.

The independent variable was the **sidebar option** used (About, Feedback, or Help), and the dependent variable was the **error minimization effectiveness**, measured through error counts and participant ratings on a Ordinal scale. Participants completed tasks requiring the use of each option, with task start and end points logged automatically.

The tasks were conducted **physically** in an **unmoderated** environment to minimize researcher influence while ensuring participants had clear instructions. Data was collected using error self-reporting through post-test surveys, where participants reflected on any errors encountered during the tasks. These surveys captured quantitative data on user ratings for perceived effectiveness and engagement with the sidebar options. The within-subjects design exposed each participant to all three options, ensuring comprehensive data collection on the comparative effectiveness of the sidebar features, reducing variability caused by individual differences, and enabling a more direct comparison of the options' effectiveness in minimizing errors.

### 4.1.4 Observational Design for Research Question 4 (Shanuka)

This research question was addressed through an **within-subjects design**, as it aimed to assess the correlation between the time spent learning about add-ons and the time spent collaboratively working on assignments in Google Docs.

The independent variable was the **time spent studying the add-ons**, and the dependent variable was the **time spent working collaboratively on assignments**. Both variables were measured using a ratio scale (in minutes), ensuring precision and comparability.

Participants were tasked with learning to use a specific add-on through tutorials before collaboratively completing an assignment in Google Docs that required the add-on. Screen recording tools captured timestamps for both activities, providing accurate data for analysis. This unmoderated approach reduced observer influence and allowed participants to work naturally.

The observational design effectively captured the relationship between the two variables, providing insights into the learning curve and its impact on collaborative work efficiency.

**Summary Table**

| Research Question | Experimental Design | Independent Variable | Dependent Variable | Participants' Tasks |
|---|---|---|---|---|
| How long does it typically take a student with prior knowledge to add an add-on compared to a student without previously acquired knowledge? (Bimandee) | Between-Subjects Design | Prior knowledge of adding add-ons | Time taken to add an add-on (in seconds) | Add a predefined add-on to the workspace. |

| | | | | |
|---|---|---|---|---|
| Do the sidebar add-ons' preventative strategies ("About," "Feedback," "Help") differ significantly in their ability to assist students in minimizing errors when using particular add-ons? (Kesavan) | Within-Subjects Design | Sidebar option ("About," "Feedback," "Help") | Error minimization effectiveness (error count/Likert scale) | Perform tasks requiring use of one of the sidebar options ("About," "Help," or "Feedback"). |
| What effects does the "Form Builder Plus" add-on have on students' workflow when they switch from Google Docs to Google Forms? (Naflan) | Within-Subjects Design | Use of the add-on | Workflow engagement (task time, interactions) | Complete tasks with and without the add-on while transitioning between Google Docs and Forms. |
| Does the amount of time students spend studying the add-ons and the amount of time they spend working together on Google Docs assignments correlate? (Shanuka) | Within-Subjects Design | Time spent studying add-ons | Time spent collaborating on assignments | Study a specific add-on and complete an assignment collaboratively in Google Docs. |

***Table 7 Summary Table for Experimental Design***

## 4.2 Procedure & Planning

This section outlines the step-by-step procedures, planning, and experimental setups used for data collection corresponding to each research question. For consistency, task instructions, timings, and processes were standardized, ensuring data validity and reliability. The testing sessions were conducted either as in-person moderated or unmoderated tasks, depending on the nature of the study. Each research question had a specific experimental design tailored to address its hypothesis. The test duration varied across studies but included time for pre-task instructions, in-task activities, and post-task feedback. Task instruction sheets and related materials are referenced in the appendices to provide further details about the task plans and procedures.

### 4.2.1 Independent-Sample Test – In-Person Unmoderated Task

**Student Name:** Bimandee Premarathna

We addressed the research question, "How does the time taken by a student with prior knowledge compare to one without previously acquired knowledge to add an add-on?", through an **in-person unmoderated approach**. It means the tasks were performed within a controlled lab environment in real-time with less chance for distraction.

**Procedure:**

Participants were divided into two groups, one with prior knowledge of adding add-ons and one without. The session started with a brief **pre-test survey (4–5 minutes)** to confirm participants' prior knowledge. During the **in-test phase (20–30 minutes)**, participants were instructed to add a predefined add-on to Google Workspace. Using the "Form builder for google sheets" add-on, participants had to make a google form from the data provided in the google sheet. After that, the time to do this was recorded with the help of a timer. Then, a **post-test survey of 5-6 minutes** was conducted to gather qualitative feedback on task difficulty and experience.

**Justification:**

The unmoderated setup was chosen for the sake of time efficiency; participants were able to complete set tasks on their own, with the need for intervention limited. Flexibility that was allowed coupled with consistency from detailed instructions and self-reported task completion times reduced outside influences.

**Data Collection Tools and Materials:**

1. Timer.
2. Pre-test and post-test survey forms.
3. Detailed task instructions (**Appendix II. Task Instruction Sheet**).

### 4.2.2 Paired-Sample Test – In-Person Moderated Task

**Student Name:** Naflan Niyas

For the research question, "What effects does the 'Form Builder Plus' add-on have on students' workflow when they switch from Google Docs to Google Forms?", a paired-sample design was implemented using an in-person moderated approach.

**Procedure:**

Participants performed two tasks. First, they built a form using Google Docs and transitioned to Google Forms with the aid of the “Form Builder Plus” add-on. In the second task, they repeated the same process without the add-on. Engagement metrics such as task completion time, interaction count, and post-task survey ratings were recorded. Each task took 15–20 minutes, with an additional **10 minutes for pre-task instructions and post-task feedback**. The total session duration was approximately 45 minutes per participant.

**Justification:**

An in-person moderated design ensured that participants could consistently complete both tasks while mitigating potential errors. Moderation provided the opportunity to standardize instructions and observe participants’ engagement levels in real time.

**Data Collection Tools and Materials:**

1. Task instructions sheet (Google Document) and post-task surveys.
2. Task setup in Google Docs and Google Forms (**Appendix II. Task Instruction Sheet)**.

### 4.2.3 ANOVA – In-Person Unmoderated Task

**Student Name:** Kesavan Anbucvhelvan

To answer the research question, "Do the sidebar add-ons' preventative strategies ('About,' 'Feedback,' 'Help') differ significantly in their ability to assist students in minimizing errors when using particular add-ons?", an unmoderated task design was employed.

**Procedure:**

Participants completed three tasks using a predefined add-on, each requiring interaction with one sidebar option: “Help,” “About,” or “Feedback.” For each task, the number of errors was recorded along with the perceived usefulness ratings collected via a post-task survey. Instructions were provided in written format, and participants completed the tasks independently over a 45-minute session. **Each task took approximately 10–15 minutes** to complete. Then, a **post-test survey of 5-6 minutes** was conducted to gather qualitative feedback on task difficulty and experience.

**Justification:**

An unmoderated approach was selected to minimize observer influence, allowing participants to interact naturally with the sidebar options. The structured environment ensured data consistency while avoiding the logistical complexities of moderation.

**Data Collection Tools and Materials:**

1. Error Self-Reporting in Post-Test Surveys.
2. Written instructions and post-task questionnaires (**Appendix II. Task Instruction Sheet)**.

### 4.2.4 Pearson Correlation – In-Person Unmoderated Task

**Student Name:** Shanuka Karunarathne

For the research question, "Does the amount of time students spend studying the add-ons and the amount of time they spend working together on Google Docs assignments correlate?", an in-person unmoderated setup was implemented.

**Procedure:** Participants were divided into two groups and assigned different tutorials for learning the Google Docs "Table of Contents" add-on:

1. Group 1: Received a **2-minute** tutorial that gives a quick overview of the add-on's basic features.
2. Group 2: Received a **5-minute** tutorial with more detailed instructions and additional tips.

**Steps:**

1. Participants completed the tutorial individually to learn the add-on, with the time spent on this activity recorded.
2. They were working in groups on a predefined Google Docs assignment, which had to be completed with the add-on. Tasks involved creating headings and inserting a structured Table of Contents.
3. Time spent on both learning and collaborative tasks was recorded by using screen recording and time-tracking software.

**Session Duration:**

1. Learning Phase: 2–5 minutes depending on the group
2. Collaborative Phase: Approximately 20 minutes to complete the assignment.
3. Each session lasted about 30 minutes.

**Justification:**
Unmoderated sessions allowed participants to work at their own pace, ensuring that the collected time data accurately reflected their natural workflow. Clear instructions minimized confusion, ensuring precise data recording.

**Data Collection Tools and Materials:**

1. Timer.
2. Tutorial materials and assignment instructions (**Appendix II. Task Instruction Sheet).**

### 4.2.5 5. Participants Information Sheet

A participant information sheet and a consent form were developed to support the recruitment process. The information sheet aimed to provide participants with a clear understanding of the study's guidelines, expectations, the nature of the data to be collected, its intended use, and the measures in place to ensure confidentiality. The consent form was designed to confirm that participants fully understood the procedures involved in the study and to obtain their explicit permission to take part in the usability testing. Both the participant information sheet and the consent form used during recruitment are available in **Appendix I. Participant Information Sheet**

## 4.3 Roles & Responsibilities

This section outlines the roles and responsibilities of all members involved in the usability test. It details the instructions provided to moderators, the contributions of each group member, and the preparations made to ensure the usability sessions were conducted effectively. For comprehensive guidelines provided to the moderators, please refer to upcoming headings.

### 4.3.1 Independent-Sample Test – Roles & Responsibilities

**Student Name:** Bimandee Premarathna

As the moderator of the usability test of a controlled lab. My work was welcoming participants, clearly explaining tasks, and showing how to use the "Form builder for

google sheets" add-on, while following along with their progress to ensure that everything was as correct and consistent as possible.

**My Contributions (as a Moderator and Facilitator):**

1. I moderated an Independent-sample test session and took test participants through a pre-test, in-test, and post-test procedure.
2. I ensured that activities were implemented based on the planned structure and observed the participants without influencing their performance.
3. Time measurement data collection and analysis were done to address the research question.

**Instructions I Followed as the Moderator:**

1. Welcomed participants and explained the objectives and procedures of the test.
2. Ensured participants completed the pre-test survey to confirm prior knowledge.
3. Made sure each participant was timing the duration of completion of the task.
4. Administered the post-test survey to gather qualitative feedback on task difficulty.
5. Maintained a neutral environment and avoided influencing the participants in their performances

**Planning for the Usability Session:**

1. I reserved a quiet lab environment to minimize distractions.
2. I made sure the participants' laptops are pre-configured with Google Workspace tools required for the test.
3. Participants were recruited and scheduled for individual sessions over three days.
4. I prepared a document of task instructions, a sheet where the question data were added and a google form for pre-test surveys, and post-test surveys.

### 4.3.2 Paired-Sample Test – Roles & Responsibilities

**Student Name:** Naflan Niyas

For my study, I acted as the moderator during the in-person sessions, where I guided participants in performing tasks with and without the "Form Builder Plus" add-on. My primary focus was to ensure that the tasks were executed consistently and that engagement data was recorded accurately.

**My Contributions (as a Moderator and Facilitator):**

1. I managed all test sessions, providing participants with clear instructions and facilitating post-task surveys.
2. I monitored participant engagement during task execution to ensure data reliability.
3. I analysed collected engagement data to evaluate the impact of the add-on.

**Instructions I Followed as the Moderator:**

1. Introduced the study to participants and explained the tasks to be completed.
2. Assisted participants with setting up the "Form Builder Plus" add-on during the task.
3. Tracked participant engagement metrics and noted observations.
4. Distributed and collected post-task surveys to gather feedback on their experiences.

**Planning for the Usability Session:**

1. I reserved a room at the university library that provided a distraction-free environment.
2. Since this task requires participants to use a laptop or desktop to complete it, along with access to Google Docs and Forms, participants used their own devices during the test.
3. Participants were scheduled in advance, and the testing spanned two days.
4. Task-specific instructions and feedback forms were sent via WhatsApp and email for use during the sessions.

### 4.3.3 ANOVA Test – Roles & Responsibilities

**Student Name:** Kesavan Anbuchelvan

I conducted my usability test as an unmoderated session. My role was to prepare participants for the task by providing clear, structured instructions and ensuring the data was accurately recorded through automated tools.

**My Contributions (as a Moderator and Facilitator):**

1. I created and distributed detailed task instructions to participants in advance.
2. I ensured that participants completed their tasks independently and submitted the required data.
3. I analysed error-tracking data and participant feedback for statistical comparisons.

**Instructions I Followed as the Facilitator:**

1. Sent participants a participant information sheet and instructions for the test.
2. Verified participants' access to the required tools and materials.
3. Followed up to ensure tasks were completed within the designated timeframe.

**Planning for the Usability Session:**

1. My study did not require a physical location since it was unmoderated.
2. I provided participants with an online task environment pre-configured with sidebar options ("About," "Help," "Feedback").
3. Participants were recruited virtually, and I monitored task completion progress over one week.
4. Automated tools, including screen recording and error-tracking software, were used for data collection.

### 4.3.4 Correlational Test – Roles & Responsibilities

**Student Name:** Shanuka Karunarathne

For my usability test, I facilitated unmoderated sessions where participants independently completed tasks related to studying an add-on and working collaboratively on Google Docs assignments. My role focused on ensuring participants were equipped with the necessary information and that data collection was precise.

**My Contributions (as a Moderator and Facilitator):**

1. I prepared and distributed task instructions and tutorials for participants.
2. I ensured participants used the screen recording tool correctly to log their activity.
3. I collected and analysed time-stamped data to determine correlations between study time and collaborative work time.

**Instructions I Followed as the Facilitator:**

1. Provided participants with a participant information sheet and step-by-step task instructions.
2. Offered guidance on using the screen recording tool.
3. Ensured that participants submitted their recordings and any additional data within the set timeline.

**Planning for the Usability Session:**

1. My study was conducted remotely and did not require a physical location.
2. Tutorials and pre-configured environments were provided to participants online.
3. Follow-ups were scheduled to ensure data submission and resolve any participant issues.
4. Data collection tools included screen recording software and automated tracking systems.

Each moderator's specific instructions and guidelines can be found in **Appendix I. Participant Information Sheet**

## 4.4 Participant Profile and Recruitment

### 4.4.1 Independent-Sample Test – Participant Profile and Recruitment

**Student Name:** Bimandee Premarathna

**Participant Profile:**

The participants were university students aged 18–25, with varying levels of familiarity with adding add-ons in Google Workspace. Group 1 consisted of participants with prior experience, while Group 2 included those without any experience. This distinction was essential to investigate the research question on the efficiency differences between these two groups.

**Recruitment Methods:**

Participants were recruited using a combination of email invitations and social media posts targeted at university groups. Flyers with test details were also displayed on campus notice boards. These recruitment materials clearly outlined the study objectives, eligibility criteria, and steps for signing up. Participants voluntarily registered via an online form.

Evidence of the recruitment process, including emails and flyers, is attached in Appendix III. Participant Recruitment Evidence **Appendix III. Participant Recruitment Evidence**

### 4.4.2 Paired-Sample Test – Participant Profile and Recruitment

**Student Name:** Naflan Niyas

**Participant Profile:**

The participants were university students and young professionals aged 18–26 who frequently used Google Docs and Google Forms for academic or professional tasks. All participants had basic knowledge of third-party add-ons. This demographic was chosen to evaluate how transitioning between Google Docs and Google Forms with the "Form Builder Plus" add-on impacted workflow engagement.

**Recruitment Methods:**

Participants were recruited using email campaigns and posts in online communities

(WhatsApp) related to productivity and technology. Recruitment evidence, including sample emails and WhatsApp messages, is included in **Appendix III. Participant Recruitment Evidence.**

### 4.4.3 ANOVA Test – Participant Profile and Recruitment

**Student Name:** Kesavan Anbuchelvan

**Participant Profile:**

The participants included undergraduate students aged 18–24 with some prior experience using Google Workspace add-ons. They were required to have a basic understanding of task completion using the "About," "Help," and "Feedback" sidebar options. This profile was critical to evaluating how effectively these features helped minimize errors.

**Recruitment Methods:**

Participants were recruited through university bulletin boards, an online portal for student activities, and word-of-mouth referrals. Recruitment posters were distributed via social media platforms and student groups, ensuring a broad reach. The recruitment materials specified the study's purpose, eligibility criteria, and how to participate.

Supporting evidence for participant recruitment, such as posters and online posts, can be found in Appendix III. Participant Recruitment Evidence **Appendix III. Participant Recruitment Evidence**

### 4.4.4 Correlational Test – Participant Profile and Recruitment

**Student Name:** Shanuka Karunarathne

**Participant Profile:**

Participants were students aged 18–26 with a mix of experience levels in using Google Workspace add-ons. They were selected based on their familiarity with

studying tutorials and collaborating on group assignments. This profile was crucial for analyzing the correlation between study time and collaborative task time.

**Recruitment Methods:**
The recruitment process involved distributing emails to student mailing lists and promoting the study through online education forums. Social media platforms were also used to share a digital flyer with a registration link. The recruitment materials provided clear instructions and details about the study objectives.

Recruitment evidence, including screenshots of emails and online posts, is attached in Appendix III. Participant Recruitment Evidence **Appendix III. Participant Recruitment Evidence.**

## 4.5 Data Collection Methods

This section explains the data collection process, including the tools used, the type of data collected, and the study conditions (moderated or unmoderated, in-person or remote).

### 4.5.1 Independent-Sample Test – Data Collection Methods

**Student Name:** Bimandee Premarathna

**Data Collection Process:**
Data collection was done in person in the controlled lab environment to minimize chances of error and other forms of distraction. The study was unmoderated, and I was the session moderator. My role was to guide the participants and give the instructions to the participants, ensure adherence to the task instructions, and record the time taken for the execution of the task correctly.

**Types of Data Collected:**

1. **Quantitative Data:** include the time of participants in completing the task in seconds.
2. **Qualitative Data:** Feedback from pre-test and post-test questionnaires regarding participants' experience and task difficulty.

**Tools Used:**

1. A digital timer to measure task completion time.
2. Pre-test and post-test survey forms to gather qualitative data.

**Location and Setup:**

The experiment was done in a computer lab at the IIT Computing School. Students used their laptops, already loaded with Google Workspace, so that the testing environment would be just the same.

### 4.5.2 Paired-Sample Test – Data Collection Methods

**Student Name:** Naflan Niyas

**Data Collection Process:**

The study was conducted **in person** at the **IIT Computing School** in a controlled lab setting to ensure consistency. It was **moderated**, and I acted as the facilitator. My role included giving task instructions, observing participant behavior, and answering any clarifying questions.

**Types of Data Collected:**

1. **Quantitative Data:** Engagement metrics, such as time spent on tasks and the number of interactions.
2. **Qualitative Data:** Post-task survey ratings on a Likert scale for perceived engagement and workflow efficiency.

**Tools Used:**

1. Engagement tracking tools to monitor interactions (Since it was a physical session, video recordings were captured to evaluate participant engagement).
2. Post-task survey forms for feedback.

**Location and Setup:**

The study was conducted in a computer lab at the IIT Computing School, where students used their laptops, which were preloaded with Google Workspace to ensure a standardized testing setup.

### 4.5.3 ANOVA Test – Data Collection Methods

**Student Name:** Kesavan Anbuchelvan

**Data Collection Process:**

The study was conducted **in person** at the **IIT Computing School** and was **unmoderated** to reduce researcher influence and ensure natural participant behavior. Participants were provided with clear, written task instructions and completed the tasks independently.

**Types of Data Collected:**

1. **Quantitative Data:** Error counts during task completion and Likert scale ratings for perceived usefulness.
2. **Qualitative Data:** Feedback on ease of use for the "Help," "About," and "Feedback" sidebar options.

**Tools Used:**

1. Screen recording software to log task interactions and error counts.
2. Post-task surveys to collect qualitative feedback.

**Location and Setup:**

The study was conducted in a computer lab at the IIT Computing School, where students used their laptops, which were preloaded with Google Workspace to ensure a standardized testing setup.

### 4.5.4 Correlational Test – Data Collection Methods

**Student Name:** Shanuka Karunarathne

**Data Collection Process:**

The study was **in person** at the **IIT Computing School** and **unmoderated** to allow participants to complete tasks at their own pace. They were provided with comprehensive instructions to follow independently.

**Types of Data Collected:**

1. **Quantitative Data:** Time spent studying the add-on and time spent collaborating on the assignment (measured in minutes).
2. **Qualitative Data:** Observations from post-task feedback about the learning experience.

**Tools Used:**

1. Timer.
2. A post-task feedback form for qualitative data.

**Location and Setup:**

The study was conducted in a computer lab at the IIT Computing School, where students used their laptops, which were preloaded with Google Workspace to ensure a standardized testing setup.

## 4.6 Research Instruments (materials) and Equipment

Although the authors decided on using both moderated and unmoderated testing for this study, it was crucial to select a suitable study method to ensure that the data gathered would keep continuing to be valid and unbiased.

For this usability testing, we utilized a combination of tools and materials to ensure accurate data collection and session recording. The primary prototype used was the Google Workspace add-on, designed to evaluate its impact on participants. To gather feedback before and after testing, we created online questionnaires, including pre-test and post-test forms. The pre-test questionnaire collected demographic information and assessed participants' familiarity with Google Workspace, while the post-test questionnaire, divided into four sections, evaluated each participant's study method. Tasks were conducted using participants' laptops in a controlled lab environment, with high-resolution mobile phone cameras used to capture participant interactions for additional qualitative insights. Clear, step-by-step written instructions were provided to ensure consistency across sessions. These tools and materials were selected for their relevance to the research objectives. The questionnaire could be found under

## **Appendix IV. Data Collection** Material.

Down below, a table has been provided with all the questions which could be found in the questionnaires, and the reasoning behind including them in the questionnaires.

| **Pre-Test Questions** | |
|---|---|
| **Questions** | **Purpose of the question** |
| **Q1**. Name | To gather basic demographic information about the participant. |
| **Q2**. Age | To understand the age group of participants, which may help in analyzing user behaviour or preferences. |
| **Q3**. What is the degree program you are following? | To assess the participant's academic background, which could influence their familiarity and comfort with digital tools. |
| **Q4.** Are you a frequent user of Google Workspace Application (Google Doc, Slides, Sheets, Form)? | To assess the participant's level of experience with Google Workspace, which could impact their interaction with the add-on. |
| **Q5**. Did you know that Google Workspace includes a sidebar located on the right side? | To gauge the participant's knowledge of Google Workspace's features and whether they are familiar with the interface elements relevant to the add-on. |
| **Q6**. Overall, how would you rate the usage of add-ons in workspace applications? | To understand the participant's perception and experience with add-ons in Google Workspace applications. |
| **Q7**. Do you think using these add-ons | To assess the participant's opinion on the |

| | |
|---|---|
| helps you work more efficiently and follow a proper workflow? | effectiveness of add-ons in improving workflow efficiency. |
| **Q8**. How easy is it to access add-ons through extensions tab compared to sidebar add-ons? | To gauge participant's familiarity with accessing add-ons and their ease of use across different methods. |
| **Q9**. Did you encounter any challenges when installing third party add-ons? | To identify any potential obstacles participants might face when installing add-ons, which could affect the overall usability experience. |
| **Q10**. If yes, what challenges did you encounter? | To get specific feedback on the challenges encountered during the installation process, which may inform areas of improvement for the add-on. |
| **Q11**. Are you willing to participate and provide feedback in a follow-up usability test in which you will complete a set of tasks in Google Add-ons and answer some interview questions? | To confirm the participant's willingness to participate in the full usability testing session and provide feedback for further analysis, |

*Table 8 Pre-Test Question Table*

| **Post- test Questions (Naflan)** | |
|---|---|
| **Moderated (Within Group)** | |
| **Study Method: Paired Sample Test** | |
| **Usability Metrics: User Engagement** | |
| **Questions** | **Purpose of the question** |

| | |
|---|---|
| **Q1**. Have you used any add-ons for creating Forms before? | To assess participants' prior experience with add-ons, which could influence their engagement and ease of use with the new tool. |
| **Q2**. Does the tool keep you interested in completing the task? | To measure user engagement by assessing how effectively the add-on holds the participant's attention during the task. |
| **Q3**. Do you feel that this add-on is saving your time compared to traditional copy paste methods? | To gauge how the participant perceives the add-on to be in comparison to traditional methods, which contributes to their engagement and overall satisfaction. |
| **Q4**. How smooth and enjoyable was the process of migrating content using the add-on compared to the traditional copy-pasting method? | To assess user engagement by evaluating the perceived ease and enjoyment of using the add-on compared to a manual method. |
| **Q5**. Does using the Form Builder Plus add-on make the process more engaging and simplify completing the form-building task compared to using the traditional copy-paste method? | To directly assess whether the add-on enhances user engagement and simplifies the task, comparing it with traditional methods. |
| **Q6**. If you would like to submit your results, please provide the form links here | To collect data for follow-up or validation, and to see if users are motivated enough to share their results after completing the task, which also ties into engagement. |

***Table 9 Paired sample test post test questions and purposes***

| Post- test Questions (Kesavan) | |
|---|---|
| **Unmoderated (Within Group)** | |
| **Study Method: ANOVA Test** | |
| **Usability Metrics: Error Tolerance** | |
| **Questions** | **Purpose of the question** |
| **Q1**: How helpful was the **"About"** option for completing Task 1? (Rate from 1–5) | To assess the perceived usefulness of the **"About"** option in assisting participants with completing Task 1. |
| **Q2**: Did the **"Help"** option provide clear steps for resolving the issue in Task 2? (Yes/No) | To evaluate whether the **"Help"** option provides sufficient clarity and actionable guidance for troubleshooting issues in Task 2. |
| **Q3**: Were you able to submit feedback without errors in Task 3? (Yes/No) | To determine the functionality and error-friendliness of the "Feedback" option during Task 3. |
| **Q4**: Did any of the options feel confusing or difficult to use? (Options provided) | To identify specific sidebar features ("About," "Help," or "Feedback") that may cause confusion or usability challenges for participants. |
| **Q5**: On a scale of 1–5, how effective was the "About" option in helping you complete your task? | To quantify the effectiveness of the "About" option in task completion from the participant's perspective. |
| **Q6**: On a scale of 1–5, how effective was the "Help" option in troubleshooting your problem? | To quantify the troubleshooting effectiveness of the "Help" option as perceived by participants. |
| **Q7**: On a scale of 1–5, how effective was the "Feedback" option for | To measure the effectiveness of the "Feedback" option in successfully |

| | |
|---|---|
| submitting your input? | enabling participants to provide input. |
| **Q8**: Which of the three options was the most helpful? ("About," "Help," "Feedback") | To determine participant preference for the most effective sidebar option among the three. |
| **Q9**: Did using the sidebar options minimize the number of errors compared to what you would expect without assistance? (Yes/No) | To evaluate whether the sidebar features collectively contributed to error reduction during the tasks. |

***Table 10 ANOVA test post test questions and purposes***

| Post- test Questions (Bimandee) | |
|---|---|
| **Unmoderated (Between Group)** | |
| **Study Method: Independent Sample Test** | |
| **Usability Metrics: Efficiency** | |
| **Questions** | **Purpose of the question** |
| Q1. Have you made a google form from a spreadsheet using Add-ons before? | The question measures pre-experience of the participant with similar tasks that may affect performance and insight during the experiment. It puts into perspective task completion times and usability ratings, which may have been performed faster by experienced users and found easier. |
| Q2. What was your start time? | This will help in gauging the exact time participants started the task. This, coupled with the end time, provides a metric for efficiency, which is the time taken to complete the task. |
| Q3. What was your end time? | Together with the question of the start time, this has been used to calculate the total duration of the task-a very important metric to establish the efficiency of the process. |
| Q4. Did you complete the tasks successfully? | This determines the effectiveness of the tool and the instructions provided. This |

| | |
|---|---|
| | identifies whether the participants encountered serious challenges that prevented task completion. |
| Q5. How easy was it to understand the instructions for adding the add-on? | This assesses the usability of the instructions in determining whether they were clear and accessible. This would provide insight into whether any revision was necessary to make the content more understandable. |
| Q6. How easy was it to use the Form Builder add-on to create the Google Form? | The test measures perceptions of the add-on's usability, based on how intuitive the tool was to use. It serves in the evaluation of design and usability of the add-on. |
| Q7. Did you find the process efficient? | This directly captures participants' perception of efficiency to complement the objective metric of task duration. It offers qualitative feedback about whether or not the process was at the standard level or expectation with the participant's expectation of speed and ease. |
| Q8. Do you think the instructions provided were sufficient to complete the task? | This question assesses if the instructions provided sufficient support to participants in completing the task. It is useful for pointing out what might have gone wrong in the instructional design. |

| | |
|---|---|
| Q9. If you wish, upload the link of the google form you created? | This is an optional method for verifying that the task has been completed and for assessing the quality of the participant's output. This ensures that the forms were created correctly according to the requirements of the task and adds a little more depth to the question of task success. |

*Table 11 Independent Sample Test post test questions and purposes*

| **Post- test Questions (Shanuka)** | |
|---|---|
| **Unmoderated (Between Group)** | |
| **Study Method: Pearson-Correlation** | |
| **Usability Metrics: Ease of learning** | |
| **Questions** | **Purpose of the question** |
| Which tutorial did you get? | To identify which group the participant belongs to (2-minute or 5-minute tutorial) for between-group comparison. |
| How familiar were you with Google Docs add-ons before this study? | To assess participants' prior experience, which could affect learning and task performance. |
| Have you used the specific add-on provided in this study before? | To control for participants' familiarity with the specific add-on, which might |

| | |
|---|---|
| | influence usability feedback. |
| How easy was it to learn the add-on? | To measure ease of learning, one of the usability metrics for the add-on. |
| How much time do you think was necessary to fully understand the add-on? | To collect subjective perceptions about the learning process and adequacy of tutorial time. |
| How helpful was the add-on for completing the assignment? | To evaluate the add-on's effectiveness in assisting with task completion. |
| Did the add-on save you time compared to completing the task without it? | To assess task efficiency by comparing perceived time savings with and without the add-on. |
| Do you think the time spent learning the add-on affected the time spent on the assignment? | To investigate whether learning time impacts task performance, supporting the research hypothesis. |

*Table 12 Pearson-Correlation Test post test questions and purposes*

## 4.7 Pilot Testing

Pilot testing was conducted with three participants to validate the experimental setup and ensure that the procedures and tools functioned as intended. The primary goals of the pilot testing were:

1. **Validate Tools**: Ensuring that the data collection tools and platforms (e.g., timers, surveys, and Google Workspace) operated correctly and reliably.
2. **Check Task Clarity**: Verifying that the task instructions were clear and easily understandable for participants.
3. **Evaluate Usability**: Observing whether participants could perform the tasks without undue confusion or frustration.

4. **Estimate Timing**: Gauging the approximate time required to complete each phase of the study to plan sessions more effectively.
5. **Identify Challenges**: Detecting potential issues, such as technical glitches or procedural ambiguities, to refine the setup before the full study.

**Steps in Conducting Pilot Testing:**

The pilot testing began with the recruitment of three participants who met the study's profile criteria. These participants were briefed on the testing procedures and asked to perform the tasks under the same conditions planned for the main study. During the sessions, data collection tools were monitored to confirm accuracy, and participants provided real-time feedback on the clarity of instructions and usability of the setup. Observations were also made regarding the timing of each task to ensure realistic estimates for the main study.

**Adjustments After Pilot Testing:**

Based on insights from the pilot test, several refinements were made to improve the study. Task instructions were revised to enhance clarity, and minor adjustments were implemented in the data recording tools to ensure precision. Timing estimates were fine-tuned to better align with participants' actual task completion durations. Additionally, potential challenges identified during the pilot testing, such as the need for clearer guidance during transitions between tasks, were addressed to streamline the testing process.

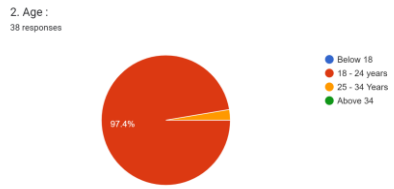
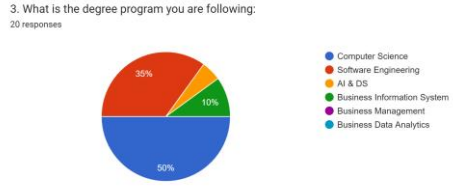
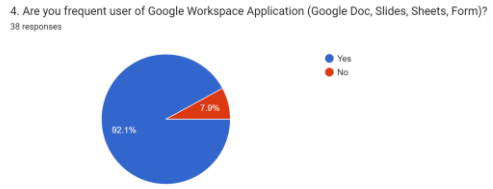
The feedback and observations from the pilot testing were instrumental in improving task instructions, fine-tuning the session structure, and addressing logistical concerns to ensure smooth execution during the actual testing phase.

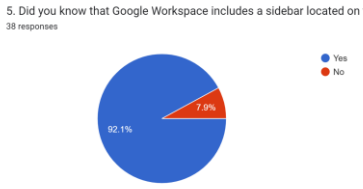
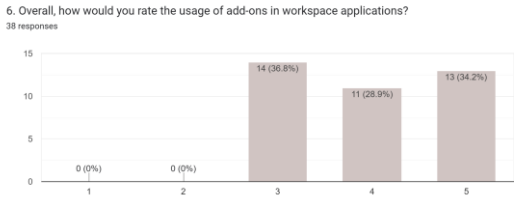
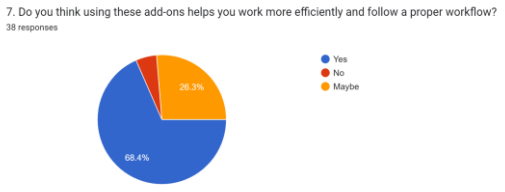
# 5. Report the study results

Following the distribution of the test questionnaire forms to the selected participants, the survey responses are addressed here and can be accessed in

Appendix VI. Raw data and evidence of test recordings **Appendix V. Test Recordings**

Below, a table is provided with all the results of the data gathered, along with an analysis of the responses.

| **Pre-Test Questions** | |
|---|---|
| **Questions** | **Analysis** |
|  | 94.9% of participants were aged 18–24, while 5.1% were aged 25–30, all being undergraduates. This indicates the test predominantly reflects the preferences and experiences of younger undergraduates, with a minor representation from older students |
|  | The participants were asked which degree program they are following. Majority of them were following Computer Science and the second highest was for Software Engineering. |
|  | Participants were asked about their familiarity with Google Workspace Applications, the product/application being examined in this research study. The majority of 92.3% participants in this research had prior experience of Google Workspace Applications. |

| | |
|---|---|
|  | Results indicate that 89.7% of participants are aware of the sidebar feature in Google Workspace, while 10.3% are not. This demonstrates that most users are familiar with this functionality, though a small portion may need guidance or education to fully utilize it. |
|  | The survey results show that 35.9% of participants rated the usage of add-ons in Google Workspace applications as neutral, 30.8% rated it as good, and 33.3% rated it as excellent. This indicates that while the majority find add-ons useful and effective, there is room for improvement to elevate the user experience for those with average satisfaction. |
|  | The survey results reveal that 66.7% of participants believe using add-ons in Google Workspace helps them work more efficiently and follow a proper workflow. Meanwhile, 28.2% were uncertain, and the remaining participants did not find add-ons beneficial. This highlights that while the majority see value in add-ons, some users may require further awareness or |

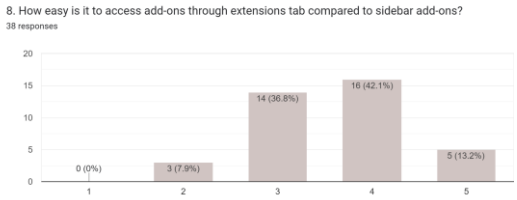
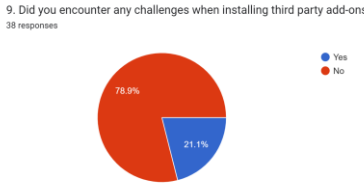
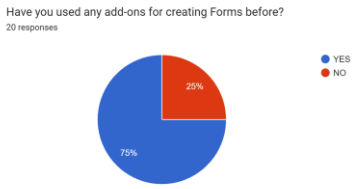
| | |
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| | training to realize their potential benefits. |
|  | The survey results indicate that 12.8% of participants find it very easy to access add-ons through the extensions tab, while 38.3% rated it as average and 41% rated it good. This suggests that while many users find it relatively easy to access add-ons through the extensions tab, there is still a significant portion that feels it could be improved for greater ease of access. |
|  | The survey results show that 79.5% of participants did not encounter any challenges when installing third-party add-ons, while the remaining participants did experience difficulties. This suggests that the majority of users found the installation process smooth. |
|  | The survey results indicate that 89.7% of participants are willing to participate in a follow-up usability test involving Google Add-ons and provide feedback, while the remaining participants declined. This high level of willingness suggests strong engagement and interest in contributing to further usability testing |

***Table 13Pre-Test Question Analysis***

| Post- Test Questions (Naflan) | |
|---|---|
| **Moderated (Within Group)** | |
| **Study Method: Paired Sample Test** | |
| **Usability Metrics: User Engagement** | |
| **Questions** | **Purpose of the question** |
|  | From the survey, over 75% of participants indicated prior experience using add-ons for creating forms, suggesting familiarity with similar tools, which may positively impact their engagement and usability during the study. |
|  | These results indicate that the majority of participants found the tool engaging and interesting, with 65% leaning toward a positive interest level (4 or 5), suggesting that the add-on successfully maintains user engagement. |
|  | 15% of participants strongly agreed, 40% agreed, and 15% remained neutral. This indicates that a majority 55% perceive the add-on as time-saving, while a smaller proportion of participants were either neutral or did not find it significantly faster. |

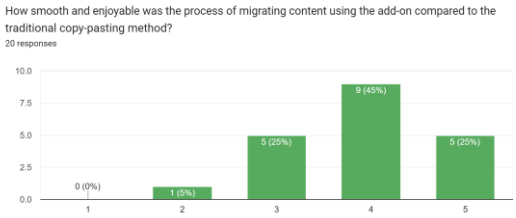
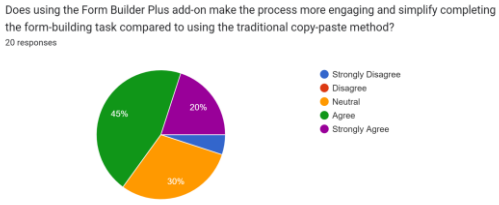
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|  | These results suggest that 70% of participants found the process to be smooth and enjoyable, highlighting the add-ons effectiveness in enhancing user experience compared to the traditional method. |
|  | The result shows that 65% of participants found the add-on both engaging and simplifying, while the remaining 30% were undecided, suggesting room for improvement in user experience for some participants. |

*Table 14 Paired Sample Test Post-Test Question Analysis*

**1. Primary Findings**

The study revealed that over 75% of participants had prior experience with add-ons, which likely reduced the learning curve and allowed them to focus on evaluating the "Form Builder Plus" add-on. A significant 65% rated their engagement with the tool positively, indicating its ability to maintain interest during tasks. Additionally, 55% perceived the add-on as time-saving compared to traditional methods, though 15% remained neutral. The process of migrating content was rated as smooth and enjoyable by 70% of participants, reflecting a positive user experience. Overall, 65% found the add-on engaging and simplifying, though 30% were undecided, highlighting opportunities for further refinement to fully engage all users.

**2. Link to Research Question and Hypothesis**

**Research Question:**

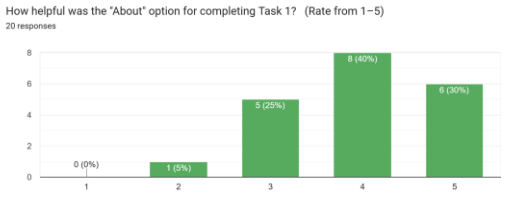
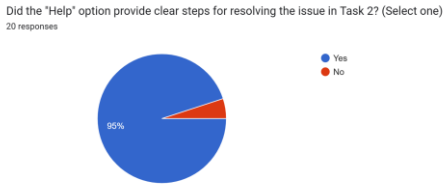
What effects does the "Form Builder Plus" add-on have on students' workflow when they switch from Google Docs to Google Forms?

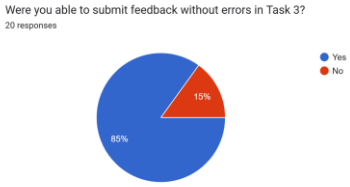
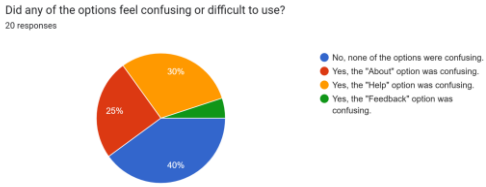
**Findings and Hypothesis Validation:**

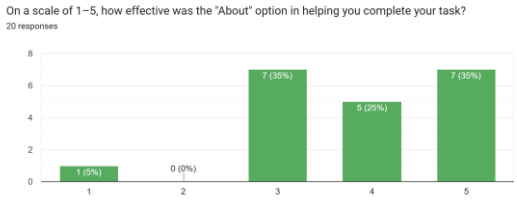
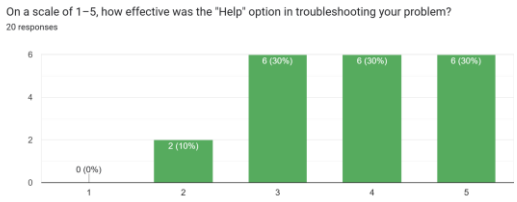
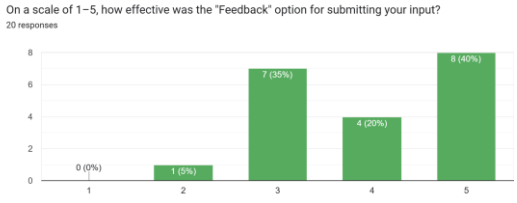
1. The data strongly aligns with the **Alternative Hypothesis ($H_1$)** that the "Form Builder Plus" add-on significantly impacts students' workflows.
2. High engagement levels (as shown in Q2, Q4, and Q5) and positive feedback on ease and efficiency indicate that the add-on enhances user workflows by making the process smoother, more enjoyable, and less time-consuming.
3. The low neutral and undecided percentages suggest that the add-on's design and functionality meet the expectations of the majority, proving its utility in enhancing engagement.

## 3. Implications for Usability Metrics

The study assessed user engagement with the "Form Builder Plus" add-on by examining how effectively it captured and sustained user interest through multiple dimensions. Feedback on interest levels showed that 65% of participants found the tool engaging, while 70% appreciated its smooth and enjoyable workflow, reflecting a positive user experience. Participants also acknowledged the add-on's efficiency, with most agreeing it saved time compared to traditional methods, highlighting its practical value. Additionally, the willingness of users to submit their completed forms demonstrated their active involvement and satisfaction with the tool. Despite these positive outcomes, approximately 30% of participants expressed neutrality or indecision, indicating opportunities for improvement. These findings suggest that while the add-on is effective for most users, enhancements in features or clearer guidance could further boost engagement across all user groups.

| Post- test Questions (Kesavan) | |
|---|---|
| **Moderated (Within Group)** | |
| **Study Method: ANOVA Test** | |
| **Usability Metrics: Error Tolerance** | |
| **Questions** | **Purpose of the question** |
|  | The responses indicate a moderate to high perception of the "About" option's helpfulness. A majority (40%) rated it 4, suggesting the feature provided adequate guidance. Another 30% rated it 5, showing that for some participants, the option was extremely effective. However, the remaining responses (25% rated 3, 5% rated 2) reveal room for improvement. The lower ratings could indicate difficulties in understanding or applying the provided information. This highlights a mixed user experience that warrants further refinement of this feature. |
|  | The overwhelming majority (95%) responded "Yes," indicating that the "Help" option effectively provided clear, actionable steps. The minimal "No" responses (5%) may stem from user-specific challenges such as difficulty |

| | |
|---|---|
| | interpreting the steps or inadequate familiarity with the interface. This demonstrates strong overall performance with minor areas for improvement. |
|  | Most participants (85%) successfully submitted feedback without errors, indicating the system's high reliability. However, the 15% who faced issues suggest the presence of occasional technical or usability challenges. The "Feedback" option's design may need fine-tuning to ensure seamless performance for all users, especially under less ideal conditions. |
|  | While 40% of participants reported no confusion with any option, 30% found the "Help" option challenging, followed by the "About" option at 25%. This indicates that despite its general effectiveness, the "Help" option might require clearer navigation or instructions. The confusion related to "About" suggests a need to simplify or reframe the information provided. The least confusion was associated with "Feedback" (5%), highlighting it as the most straightforward option. |

| | |
|---|---|
|  | The ratings show a bimodal distribution: 35% rated it 5 (very effective), but an equal percentage rated it 3, showing a moderate level of effectiveness for many. The middle rating indicates the "About" option works well for some tasks but might lack comprehensiveness or clarity for others. The feedback here underscores the need to refine the content or delivery of this feature to cater to a broader user base effectively. |
|  | The "Help" option received uniformly distributed ratings across 3 (30%), 4 (30%), and 5 (30%). This even spread suggests that while the feature is generally effective, its utility varies based on user experience or the complexity of the problem. The absence of ratings at 1 and a low score at 2 (10%) highlights its reliability, but improvements in customization or detail could boost user satisfaction. |
|  | The "Feedback" option was rated highly effective by 40% of participants (score of 5) and moderately effective by another 35% (score of 3). A smaller percentage (20% rated 4, 5% rated 2) suggests some variability in user experiences, potentially influenced by technical factors or task-specific nuances. These |

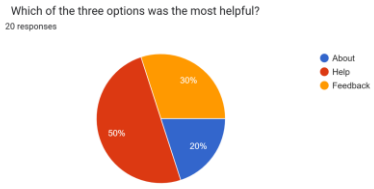
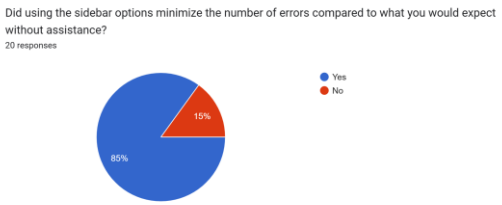
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| | results affirm its strong utility, with some minor improvements needed to address occasional inconsistencies. |
|  | The "Help" option emerged as the most helpful (50%), affirming its dominant role in supporting users. The "Feedback" option followed at 30%, while "About" lagged at 20%. These results highlight the need to enhance the "About" feature to provide clearer, more actionable guidance, ensuring its competitiveness with the other two options. |
|  | The majority (85%) agreed that the sidebar options minimized errors, reinforcing the effectiveness of these features. The 15% dissenting may have faced technical issues or limitations in the features' ability to address specific scenarios. These findings validate the importance of such tools in reducing user errors. |

*Table 15 ANOVA Test Post-Test Question Analysis*

### 1. Primary Findings

The results of the study revealed that the "Help" sidebar option was the most effective overall, as it garnered the highest number of positive responses across multiple questions. Specifically, 95% of participants reported that the "Help" option provided clear steps for resolving issues (Q2), and 50% identified it as the most helpful feature

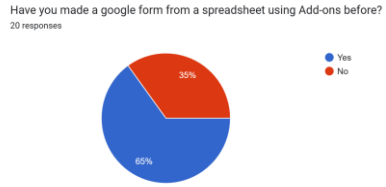
(Q8). These findings highlight its strong role in guiding users through troubleshooting tasks effectively. Similarly, the "Feedback" option demonstrated high performance, with 40% of participants rating it as very effective for submitting input (Q7). However, the "About" option received mixed feedback, with a moderate level of effectiveness reported by participants. While some users rated it positively, others found it less helpful or confusing compared to the "Help" feature. These findings indicate variability in user experiences across the three options and suggest a need for improvements in the "About" feature to enhance its usability and clarity.

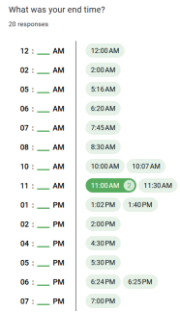
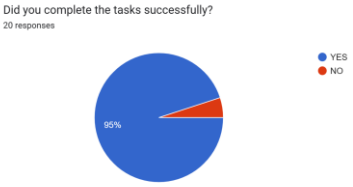
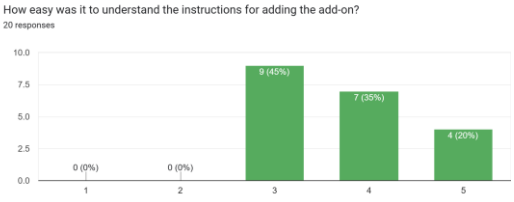
**2. Link to Research Question and Hypothesis**

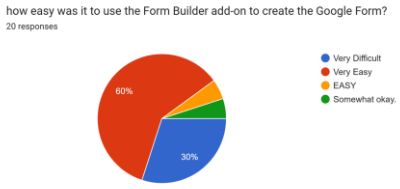
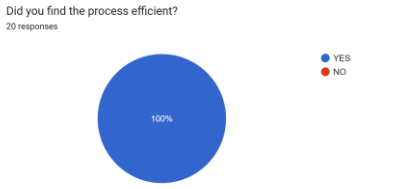
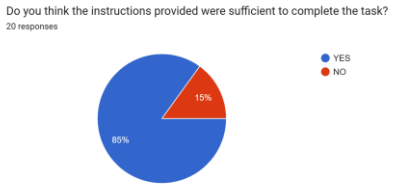
The study's findings directly address the research question by highlighting significant differences in the effectiveness of the three sidebar options in minimizing errors. The "Help" option's dominance in both clarity and overall utility supports the **Alternative Hypothesis ($H_1$)**, which posits that at least one preventative strategy differs significantly in its ability to assist users. The observed variance in ratings and effectiveness scores among the options confirms that these differences are statistically meaningful. Consequently, the **Null Hypothesis ($H_0$)** stating no significant difference among the sidebar options is rejected. These results demonstrate that while all three options contribute to error minimization, their relative effectiveness varies, with "Help" emerging as the most impactful.

**3. Implications for Usability Metrics**

The findings provide valuable insights into the usability metrics of **Error Tolerance** and **User Clarity**. In terms of **Error Tolerance**, the majority of participants (85%) agreed that the sidebar options reduced errors compared to expectations without assistance (Q9). Both the "Help" and "Feedback" options received higher ratings for their ability to facilitate error-free task completion, showcasing their reliability. For **User Clarity**, the "Help" option stood out by providing clear and actionable steps, as indicated by a 95% positive response rate for Q2. This clarity translated into efficient task completion, making the "Help" feature a critical tool for users. However, the mixed reception of the "About" option suggests that additional efforts are needed to align its content and design with user needs. These findings emphasize the importance of refining all sidebar options to ensure consistent performance and enhanced usability across the board.

| Post- test Questions (Bimandee) | |
|---|---|
| **Unmoderated (Between Group)** | |
| **Study Method: Independent Sample Test** | |
| **Usability Metrics: Efficiency** | |
| **Questions** | **Purpose of the question** |
|  | From the survey, 65% responded that they had previously made forms using add-ons, and thus, it can be assumed that the majority of the participants were familiar with similar tools. This should advantage them in understanding the task and solving it efficiently, while the 35% who did not have prior experience gave insight into how intuitive the process was for novices. |
|  | In the survey, all participants started the task at the same time, 6:10 PM, thus ensuring that the starting of the task was consistent among the participants. This coordination eliminated the potential for variability in the start times. |

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|  | It should be noted that since all respondents concluded before 6:30 PM, the task was do-able within a reasonable timescale. The consistency with which this happens suggests a clarity of the instructions in that the process was reasonably efficient for experienced and novices alike. |
|  | An impressive 95% of the participants reported success in completing the tasks, indicating that for the majority of users, the instructions given were clear and the tool was usable. The 5% that did not succeed reveal areas that could be further improved with instructions or problems that some of the participants encountered. |
|  | Most ratings fell within the middle to high range, with 45% choosing "3," 35% choosing "4," and 20% choosing "5." This suggests that, although the majority felt that the instructions were understandable, there is still further scope to increase clarity for those participants with lower technical competence. |

| | |
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|  | Correspondingly, 60% described it as "Very Easy," while another 30% considered the process "Easy." Generally, these statistics demonstrate very good reception of the add-on. The other comments being in the category "Somewhat Okay" really drive the point that designing user-friendly interactions is essential to even further smoothen the user experience. |
|  | All the participants, 100%, believed that the process was smooth and efficient, further pinpointing how effective the design of the unmoderated task was and just how functional the tool performed. Such unanimous feedback gives good backing to using the add-on and workflow designed for the study. |
|  | From the survey, 85% felt that the instructions were adequate for the performance of the task, hence the majority depended on the set guidelines. The incomprehensible 15% do bring in areas that need improvement-for instance, clarification of certain steps or even providing extra visual aids to ensure the instructions are clear enough. |

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|  | From the survey, only a few subjects decided to upload the link of the Google Form that they made. This was just an optional question in case someone needed direct evidence of tasks performed and verification that this was indeed the form produced. Since this experiment was conducted inside the lab, they could directly show us the forms created. This ensured that all forms came through reviewed and validated in real time, compensating for the limited number of uploaded links. |

*Table 16 Independent Sample Test Post-Test Question Analysis*

## 1. Primary Findings

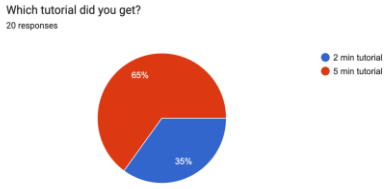
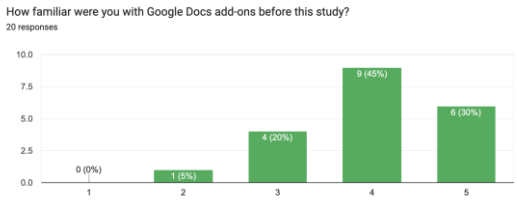
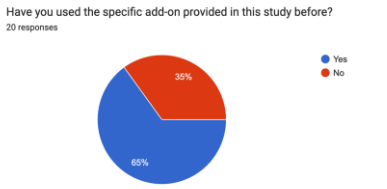
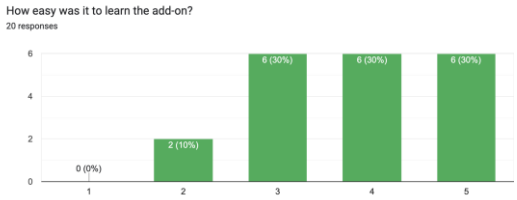
Key results from the study presented several important findings on efficiency and effectiveness related to how add-ons can be utilized to create a Google Form. The majority of test participants, 65%, indicated prior experience in using an add-on for form creation, which likely contributed to efficiently completing the task. All the participants started the task at the same time, 6:10 PM, to keep **consistency in the** start times; they all finished before 6:30 PM, demonstrating it was possible to complete it within a reasonable timeframe. In all, 95% of participants completed the task, which indicates the tool was usable and instructions clear for most users. Whereas 85% felt that the instructions were sufficient, 15% expressed difficulty and showed areas that needed further clarification. On the ease of use of the Form Builder add-on, 60% found it "Very Easy," while 100% reported that the process was efficient. These findings show the effectiveness of the tool in streamlining the form creation process. While only a few participants uploaded their form links, the real-time verification made possible by the lab setup ensured the authenticity of this result.

## 2. Link to Research Question and Hypothesis

These findings address the research question on the effectiveness and usability of the process of creating a Google Form using add-ons. The high success rate of 95%, combined with the unanimous positive feedback provided on task efficiency (100%), is supportive of the **Alternative Hypothesis ($H_1$)**, postulating that the add-on, together with the instructions, effectively facilitated the execution of tasks. The fact that they have all finished before 6:30 PM strengthens the start and end times, in fact, and thus reinforces the fact that the results are reliable, showing that the task was well-designed and achievable in the given reasonable time. The 15%, who had problems, confirmed that the **Null Hypothesis $H_0$**-possible improvements for users with less experience-will still be there. However, the overall success of the task and feedback suggest that this add-on is effective for most users.

## 3. Implications for Usability Metrics

The results also yield rich information on two key usability measures: **Efficiency** and **User Clarity**. With regard to the former, Efficiency, a unanimous response from the participants on the efficiency of the process (100%) and their completion of the task before 6:30 PM evidence the streamlined nature of the design of the tool and task. Regarding User Clarity, the majority, 85%, found the instructions adequate, while 15% struggled with them. This clearly points to a need for improvement, which could take the form of more detail in the instructions or supplementary visuals. These results support the importance of clarity regarding user experience. The regularity in start and end times, combined with real-time verification of the completion of lab tasks, speaks to the need for managing the flow of tasks and checking user output to get accurate results.

| Post- test Questions (Shanuka) | |
|---|---|
| **Unmoderated (Between Group)** | |
| **Study Method: Pearson-Correlation** | |
| **Usability Metrics: Ease of learning** | |
| **Questions** | **Purpose of the question** |
|  | 65% of the participants received the 5-minute tutorial, whereas 35% received the 2-minute tutorial. This indicates a greater proportion in the longer learning group. |
|  | Most of the participants were moderately familiar with the add-ons on Google Docs, with 45% rating their familiarity as 4 and 30% as 5, showing a higher familiarity level. |
|  | 65% of participants had prior experience with the add-on, suggesting that prior familiarity may influence ease of learning and task performance. |
|  | Most of them found the add-on easy to learn: 30% rated it 4, and another 30% rated it 5, reflecting positive feedback on ease of learning. |

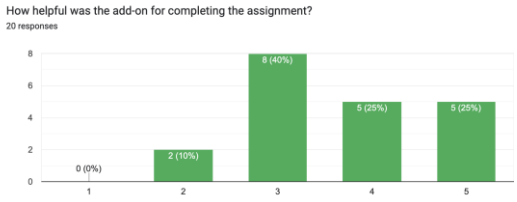
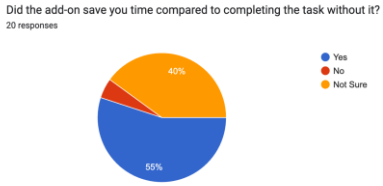
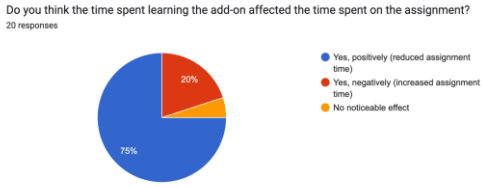
| | |
|---|---|
|  | 40% rated the add-on as moderately helpful (3), while 25% rated it very helpful (5), reflecting that opinions about its usefulness were varied. |
|  | 55% agreed that it saved time, but 40% were unsure, thus mixed perceptions on its efficiency benefits. |
|  | 75% felt that learning time positively impacted assignment time, thus supporting the hypothesis that effective learning reduces task time, though 20% noted a negative impact. |

*Table 17 Pearson-Correlation Post-Test Question Analysis*

## 1. Primary Findings

Results indicate that 65% of the sample felt a 5-minute tutorial was warranted, while 35% felt 2 minutes of a tutorial was warranted. In addition, 75% rated their familiarity with Google Docs add-ons a 4 or 5 (1 = not familiar, 5 = extremely familiar), indicating that the majority of the sample were comfortable with the platforms. 65% indicated they had used the add-on used in the study before. 60% rated how learnable it was between 4-5. However, in regards to how long it took them to learn it, the results were scattered; 1-30 minutes, which indicates that some may have taken longer than others to feel they had a grasp on it. 5% felt that learning the add-on was a learn-time saver; however, 40% were neutral about the add-on being a learn-time saver. Regardless, 75% agreed that learning how to use the add-on was a time-saver when completing an assignment, which shows that spending time learning something the proper way can increase efficiency down the line.

## 6. Ethics form

Since the start of this study, the moderators have made it very evident that all ethical rules and regulations must be adhered to in order to ensure that no problems resulting from unethical behavior occur. Therefore, "**Part- A Research Ethics Consideration Application form**" was filled out at the outset of this study, and we also made it apparent that all of its rules should be strictly adhered to. Furthermore, steps were taken to guarantee that research participants felt secure and at ease during their participation. They received information papers and were well told about all the procedures in advance, and maintaining their privacy was a top focus. They were given the opportunity to withdraw from the study at any moment if they so desired, and their permission to participate in the test was completely secured. The Ethics cover sheet along with the Part A Research Ethics Consideration Application form has been provided below in the **Appendix VII. Ethics form.**

## 7. Conclusions

As previously mentioned, we evaluated the usability of the "Google sidebar add-ons in Google Workspace Applications" with a group of 20 student participants using four different methods. These testing techniques included moderated within-group testing with tasks divided into two pieces and unmoderated between-group testing with two task sections. We methodically developed comprehensive test plans for each approach and provided participants with comprehensive documentation so they could be certain they understood every step.

We created a task plan, consent form, participant information sheet, and questionnaire in accordance with ethical guidelines and distributed them to the participants. The study required participants to complete the test materials and questionnaires within approximately one and a half hours, during which they were expected to use the materials carefully.

The collected data was then thoroughly analyzed on the basis of each of the hypotheses that had been put out. Important information will be provided by the test findings, which

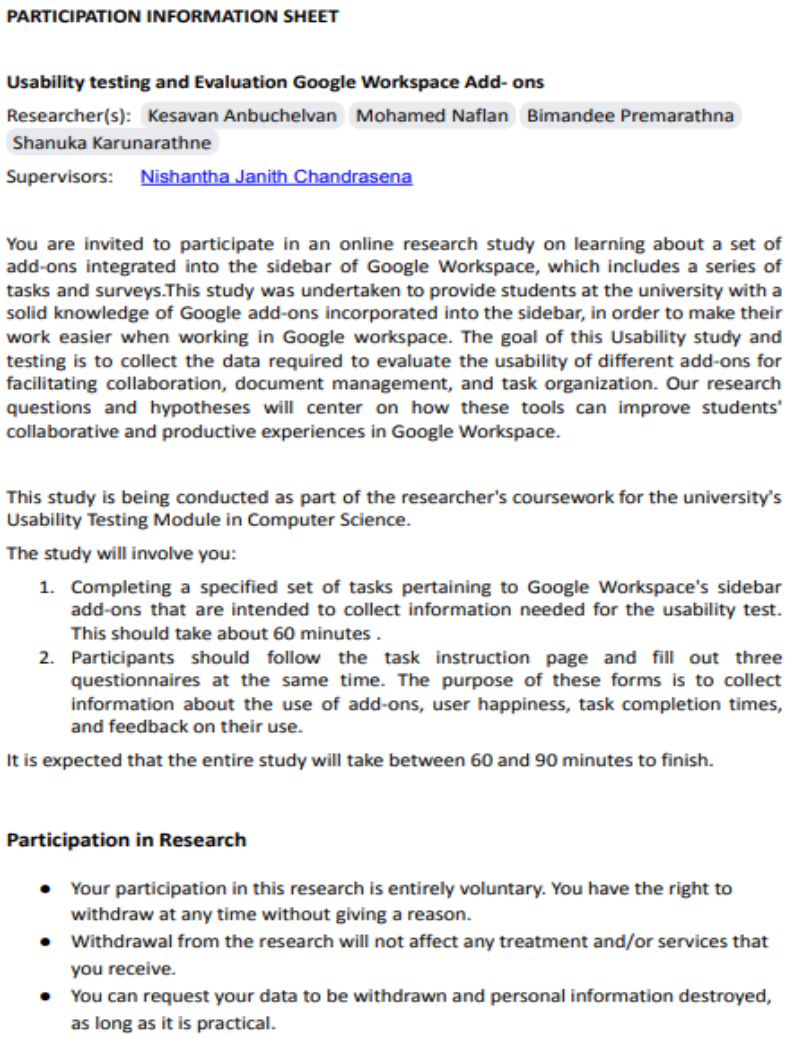
will determine whether each hypothesis is accepted or rejected. The information gathered for the coursework offers a strong foundation for assessing the suggested hypotheses. Each hypothesis was carefully examined, and the validity of each was evaluated by selecting the most suitable statistical method.

# Appendix

## Appendix I. Participant Information Sheet

Participants information sheet: [Participant Information Sheet and consent form.docx]

Consent form: [https://forms.gle/ywvhGYCZBpDyQHt7A](https://forms.gle/ywvhGYCZBpDyQHt7A)



**PARTICIPATION INFORMATION SHEET**

**Usability testing and Evaluation Google Workspace Add- ons**

Researcher(s): Kesavan Anbuchelvan Mohamed Naflan Bimandee Premarathna Shanuka Karunarathne

Supervisors: Nishantha Janith Chandrasena

You are invited to participate in an online research study on learning about a set of add-ons integrated into the sidebar of Google Workspace, which includes a series of tasks and surveys.This study was undertaken to provide students at the university with a solid knowledge of Google add-ons incorporated into the sidebar, in order to make their work easier when working in Google workspace. The goal of this Usability study and testing is to collect the data required to evaluate the usability of different add-ons for facilitating collaboration, document management, and task organization. Our research questions and hypotheses will center on how these tools can improve students' collaborative and productive experiences in Google Workspace.

This study is being conducted as part of the researcher's coursework for the university's Usability Testing Module in Computer Science.

The study will involve you:

1. Completing a specified set of tasks pertaining to Google Workspace's sidebar add-ons that are intended to collect information needed for the usability test. This should take about 60 minutes .
2. Participants should follow the task instruction page and fill out three questionnaires at the same time. The purpose of these forms is to collect information about the use of add-ons, user happiness, task completion times, and feedback on their use.

It is expected that the entire study will take between 60 and 90 minutes to finish.

**Participation in Research**

- Your participation in this research is entirely voluntary. You have the right to withdraw at any time without giving a reason.
- Withdrawal from the research will not affect any treatment and/or services that you receive.
- You can request your data to be withdrawn and personal information destroyed, as long as it is practical.

***Figure 1 Participants information sheet part 1.***

- You are not required to answer particular questions in questionnaires or interviews if you do not wish to.

**Privacy and Confidentiality**

- Your responses will normally be made anonymous and kept confidential unless you provide explicit consent to do otherwise, such as for the use of your image in photographs or video recordings.
- No individuals will be identifiable from collated data, written reports, or any publications arising from this research.

**Data Security**

- All computer data files will be encrypted and password protected. The researcher will store files securely and comply with the requirements of the Data Protection Act.
- Hard copy documents, such as consent forms and completed questionnaires, will be kept securely in a locked cupboard, preferably on university premises.
- Documents may be scanned and stored electronically to enable secure transmission to the university's secure computer systems.

**Post-Research Communication**

- If you wish, you can request information on the research results by indicating this on the consent form.
- The researcher can be contacted via email after your participation.

**Complaints**

- If you have a complaint about this research project, you can contact the project supervisor, Nishantha Janith Chandrasena via an email.

*Figure 2Participants information sheet part 2*

**CONSENT FORM**

**Title of Study: Usability testing and Evaluation Google Workspace Add- ons**

**Lead researcher:** Kesavan Anbuchelvan Mohamed Naflan Bimandee Premarathna Shanuka Karunarathne

| | | |
|---|---|---|
| I have been given the Participation Information Sheet and/or had its contents explained to me. | ☐ | ☐ |
| I have had an opportunity to ask any questions and I am satisfied with the answers given. | ☐ | ☐ |
| I understand I have a right to withdraw from the research at any time and I do not have to provide a reason. | ☐ | ☐ |
| I understand that if I withdraw from the research any data included in the results will be removed if that is practicable (I understand that once anonymised data has been collated into other datasets it may not be possible to remove that data). | ☐ | ☐ |
| I would like to receive information relating to the results from this study. | ☐ | ☐ |
| I wish to receive a copy of this Consent form. | ☐ | ☐ |
| I confirm I am willing to be a participant in the above research study. | ☐ | ☐ |
| I note the data collected may be retained in an archive and I am happy for my data to be reused as part of future research activities. I note my data will be fully anonymised (if applicable). | ☐ | ☐ |

**Participant's Name:** _____

**Signature:** _____ **Date:** _____

This consent form will be stored separately from any data you provide so that your responses remain anonymous.

_____

I confirm I have provided a copy of the Participant Information Sheet approved by the Research Ethics Committee to the participant and fully explained its contents. I have given the participant an opportunity to ask questions, which have been answered.

**Researcher's Name:** _____
**Signature:** _____ **Date:** _____

*Figure 3Consent Form*

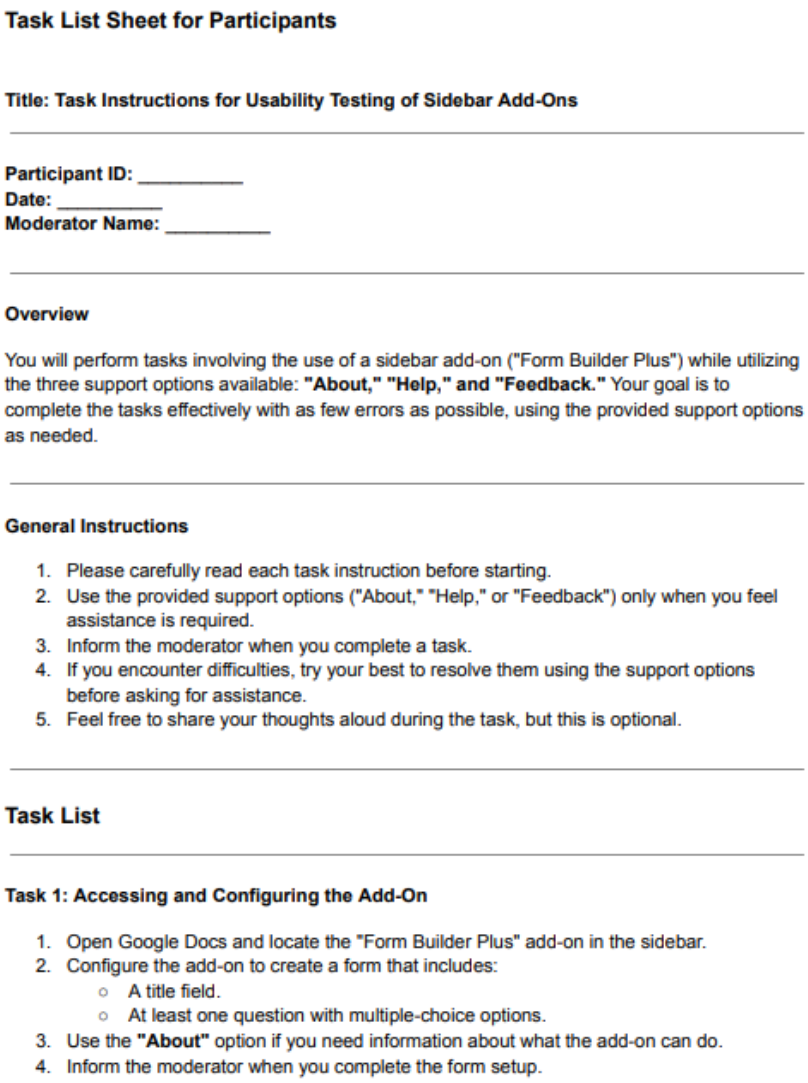
## Appendix II. Task Instruction Sheet

Anova (Kesavan) : kesavan Task List

Paired-Sample (Naflan): Instruction Sheet-Students Engagement

Independent sample test (Bimandee): Unmoderated Test - Instruction Sheet

Pearson Correlation (Shanuka) : 2-Minute Tutorial (Quick Guide)5-Minute Tutorial (Detailed Guide)



**Task List Sheet for Participants**

**Title: Task Instructions for Usability Testing of Sidebar Add-Ons**

---

**Participant ID: _____**
**Date: _____**
**Moderator Name: _____**

---

**Overview**

You will perform tasks involving the use of a sidebar add-on ("Form Builder Plus") while utilizing the three support options available: **"About," "Help," and "Feedback."** Your goal is to complete the tasks effectively with as few errors as possible, using the provided support options as needed.

---

**General Instructions**

1. Please carefully read each task instruction before starting.
2. Use the provided support options ("About," "Help," or "Feedback") only when you feel assistance is required.
3. Inform the moderator when you complete a task.
4. If you encounter difficulties, try your best to resolve them using the support options before asking for assistance.
5. Feel free to share your thoughts aloud during the task, but this is optional.

---

**Task List**

---

**Task 1: Accessing and Configuring the Add-On**

1. Open Google Docs and locate the "Form Builder Plus" add-on in the sidebar.
2. Configure the add-on to create a form that includes:
   - A title field.
   - At least one question with multiple-choice options.
3. Use the **"About"** option if you need information about what the add-on can do.
4. Inform the moderator when you complete the form setup.

*Figure 4 Anova task Instruction sheet part 1*

---

**Task 2: Troubleshooting Form Errors**

1. A sample form has already been created but contains an error (e.g., missing a required field or incorrect formatting).
2. Identify and fix the error using the **"Help"** option.
3. Inform the moderator when the error is resolved.

---

**Task 3: Providing Feedback**

1. Submit a feedback message about your experience with the add-on using the **"Feedback"** option in the sidebar.
2. Inform the moderator when the feedback submission is complete.

---

## Post-Task Questionnaire

After completing all tasks, you will be asked to rate the three support options (**"About," "Help," and "Feedback"**) based on:

1. Their **ease of use**.
2. Their **effectiveness in minimizing errors**.
3. Your overall satisfaction with each option.

---

**Moderator Notes**

- Ensure participants understand the tasks and encourage them to vocalize their thought processes.
- Observe and log any errors, confusion, or time taken for each task.
- Record any additional feedback shared by participants during the session.

---

**Thank You**

Thank you for participating in this study! Your feedback and performance will help improve the usability of the add-on.

*Figure 5 Anova task Instruction sheet part 2*

# Paired sample Test
# Student/User Engagement

## Task 1 Instructions: With form builder plus Add ons

### Step1: Install the Add-On

- Go to the Google Workspace Marketplace and install the add-on named **Form Builder Plus - Google Workspace Marketplace**

### Step2: Open the Google Doc to Work On

- Use the duplicated Assessment file as your starting document.

### Step3: Set Up the Add-On

- After installation, **refresh** your Google Docs page.
- Open the add-on from the sidebar by clicking on its icon.
- Select the option "**Simply build Form from Google Docs**" and press the **Start** button.
- Choose the "**Full Document**" option to include the entire document and click **Get.**

### Step4: Verify Parsed Content

- Check if the content from the document was parsed correctly:
  - If everything looks accurate, click YES to proceed and **create the form.**
  - If something is wrong, click Start Again to retry the parsing process.

### Step5: Create a New Form

- Under the forms section, create a **new form** and name it **Assessment**.
- Click **Create** to finalize the new form.

### Step6: Import the Form

- Once the conversion is complete, click the **Import Now** button to integrate the form into your Google Forms.

## Task 2 Instructions: Without Add-ons

### Step1: Open the Google Doc to Work On

*Figure 6 Paired-Sample task Instruction sheet part 1*

- You are provided with an Assessment file. This file contains a set of questions and their corresponding answers.
- Use the **duplicated version** of the file to ensure the original remains unchanged.

**Step2: Open a new google form**

- Go to Google Forms.
- Click the **Blank Form** option to create a new form from scratch.
- In assessment document there is couple of questions
- Copy paste each question and answer and create a form
- Rename the form to "Form 1" by clicking on the title at the top of the form.
- Save it.

After completing the above task, Please Complete the Post-test Questions under "**Section 3**" .

Link To Questionnaire : https://forms.gle/zghZLxffEUWh6Shz7 +

*Figure 7 Paired-Sample Instruction sheet part 2*

## Unmoderated Test - Instruction Sheet

**[Between-Subjects Study]**

This usability test consists of a single main task to evaluate user efficiency in adding an add-on to a workspace. Perform the task carefully, following the instructions step by step. After completing the task, answer the related questions in the provided Questionnaire.

**Estimated time:** ~10-15 minutes

**Google Sheet** -
https://docs.google.com/spreadsheets/d/18UlPPwVDItx7crW8TNBC8LWLXGs3DW6HANJpAgS6x1w/edit?usp=sharing

### Task Objective:

Evaluate the time required to add an add-on for users with and without prior knowledge of the process.

---

**Tasks**

**Task 1: Adding an Add-On to a Workspace**

1. Open the above provided Google Sheets and navigate to the "Extensions" tab in the menu bar.
2. Select "Add-ons" and click "Get add-ons."
3. In the Google Workspace Marketplace that opens, search for "Form Builder for Sheets" (or any specified add-on).
4. Click on the add-on and follow the prompts to install it.
    - **Note:** You may need to grant necessary permissions for installation.
5. Confirm the add-on has been successfully installed by checking the "Add-ons" section under the "Extensions" tab.

---

**Task 2 : Creating a Google Form from Google Sheets using the Add-on**

6. Navigate to the "Extensions" tab in the menu bar.
7. Hover over "Form Builder for Sheets" and click "Start".
8. A panel will be opened on the right side of Sheets, click "Start".
9. Choose the sheet as "Sheet 1".
10. Choose a type of Content Selection as "Range" and set the range from "A2:D5".
11. Select a pattern as "Questions and Answers".

*Figure 8 Independent Sample task Instruction sheet part 1*

12. Select Range includes header and customize the form accordingly.
    - Question at column - A
    - Image at column - (leave it blank)
    - Type at column - B
    - Description at column - (leave it blank)
    - Answer starts at column* - C
    - Answer ends at column* - C
    - Required at column - D
    - Add other at column - (leave it blank)

13. Select Delimiter.
14. Click on "GET".
15. Click on "Create" and tick sign.
16. Click on "Import Now".
17. Open the form.

---

**Completion Instructions:**

- Record the time you begin the task (when you open the "Extensions" tab) and the time you finish (when the add-on appears in the "Add-ons" section).
- Save any screenshots if required as proof of completion.
- Answer the post-task questions in the provided Questionnaire, focusing on your experience, perceived ease of use, and completion time.

---

**Post-Test Submission:**

- Submit the recorded completion time, along with any feedback, in the provided form.
- If requested, attach screenshots as proof of completion.

---

Your responses and completion time are essential to the validity of this usability study. Thank you for participating!

*Figure 9 Independent Sample task Instruction sheet part 2*

## 2-Minute Tutorial (Quick Guide)

*This tutorial focuses on the basics of installing and using the Table of Contents add-on.*

---

### Step 1: Install the Add-On

1. Open your Google Docs document.
2. Click the **Extensions** menu at the top → Select **Add-ons** → Click **"Get add-ons."**
3. In the search bar, type **"Table of Contents"** and click on the correct add-on.
4. Click the **Install** button and follow any prompts to complete the installation.

---

### Step 2: Insert a Table of Contents

1. If your document doesn't have headings, create a few sections and style them:
   - Highlight a section title → Click the **Styles** menu (Normal Text) → Select **Heading 1**, **Heading 2**, or **Heading 3.**
2. Go to **Extensions** → **Table of Contents** → Click **Insert Table of Contents.**
3. A Table of Contents will appear where your cursor is positioned.

---

### Step 3: Refresh the Table of Contents

1. If you make changes to your document (e.g., adding or removing headings), update the Table of Contents by clicking the refresh icon in the top corner of the Table of Contents box.

*Figure 10 Pearson Correlation task Instruction sheet part 1*

## 5-Minute Tutorial (Detailed Guide)

*This tutorial goes into more depth, covering customization and additional features.*

---

### Step 1: Install the Add-On

1. Open your Google Docs document.
2. Click the **Extensions** menu → Select **Add-ons** → Click **"Get add-ons."**
3. Search for **"Table of Contents"** and click on the correct add-on.
4. Click **Install** and follow the prompts to complete the installation.

---

### Step 2: Create and Style Headings

1. If your document doesn't have headings, add some:
   - Write a section title.
   - Highlight the title → Go to the **Styles** menu (Normal Text) → Select **Heading 1**, **Heading 2**, or **Heading 3.**
2. Repeat for other sections of your document.

---

### Step 3: Insert a Table of Contents

1. Position your cursor where you want the Table of Contents to appear (usually at the top or bottom of your document).
2. Go to **Extensions** → **Table of Contents** → Click **Insert Table of Contents.**
3. A Table of Contents will be generated automatically based on your headings.

---

### Step 4: Customize the Table of Contents

1. Click the Table of Contents to select it.
2. Use the toolbar that appears to choose between numbered or plain styles.
3. If you need to reorganize sections, simply edit the headings in your document; the Table of Contents will adjust.

---

### Step 5: Refresh the Table of Contents

1. Make edits to your headings, such as adding new sections or renaming existing ones.

*Figure 11 Pearson Correlation task Instruction sheet part 2*

2. Click the refresh icon in the Table of Contents box to update it automatically.

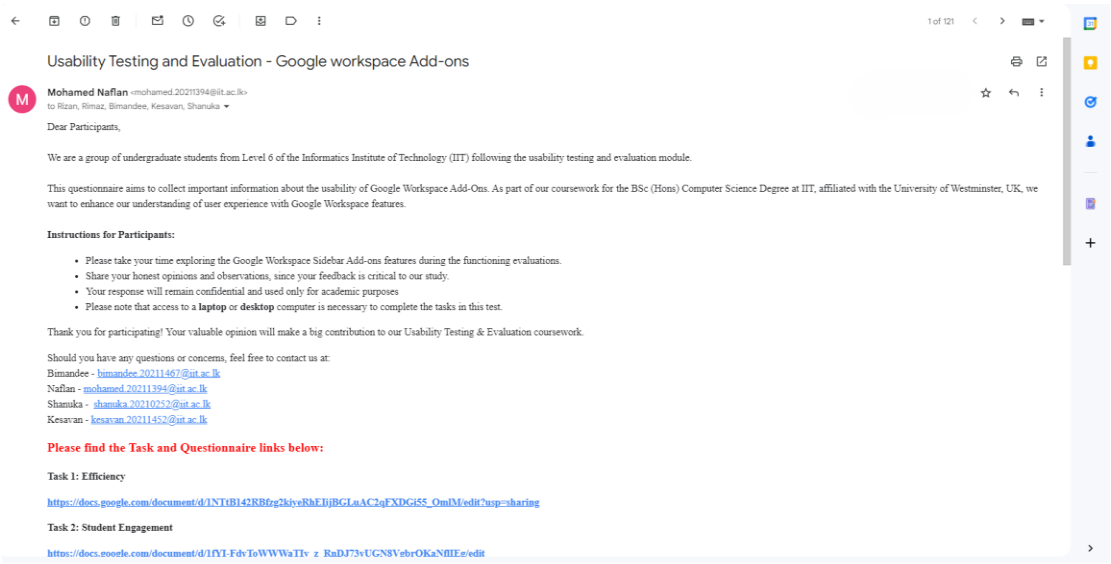
---

**Step 6: Remove the Table of Contents**

1. Click on the Table of Contents.
2. Press **Backspace** or **Delete** to remove it from the document.

*Figure 12 Pearson Correlation task Instruction sheet part 2*

# Appendix III. Participant Recruitment Evidence



Usability Testing and Evaluation - Google workspace Add-ons

**Mohamed Naflan** <mohamed.20211394@iit.ac.lk>
to Rizan, Rimaz, Bimandee, Kesavan, Shanuka

Dear Participants,

We are a group of undergraduate students from Level 6 of the Informatics Institute of Technology (IIT) following the usability testing and evaluation module.

This questionnaire aims to collect important information about the usability of Google Workspace Add-Ons. As part of our coursework for the BSc (Hons) Computer Science Degree at IIT, affiliated with the University of Westminster, UK, we want to enhance our understanding of user experience with Google Workspace features.

**Instructions for Participants:**

- Please take your time exploring the Google Workspace Sidebar Add-ons features during the functioning evaluations.
- Share your honest opinions and observations, since your feedback is critical to our study.
- Your response will remain confidential and used only for academic purposes
- Please note that access to a **laptop** or **desktop** computer is necessary to complete the tasks in this test.

Thank you for participating! Your valuable opinion will make a big contribution to our Usability Testing & Evaluation coursework.

Should you have any questions or concerns, feel free to contact us at:
Bimandee - bimandee.20211467@iit.ac.lk
Naflan - mohamed.20211394@iit.ac.lk
Shanuka - shanuka.20210252@iit.ac.lk
Kesavan - kesavan.20211452@iit.ac.lk

**Please find the Task and Questionnaire links below:**

**Task 1: Efficiency**

https://docs.google.com/document/d/1NTtB142RBfzg2kiyeRhEIijBGLnAC2qFXDGi55_OmlM/edit?usp=sharing

**Task 2: Student Engagement**

https://docs.google.com/document/d/1fYI-FdvToWWWaTIv_z_RnDJ73vUGNSVebrOKaNfIlEe/edit

*Figure 13 Participant Recruitment Evidence 1*

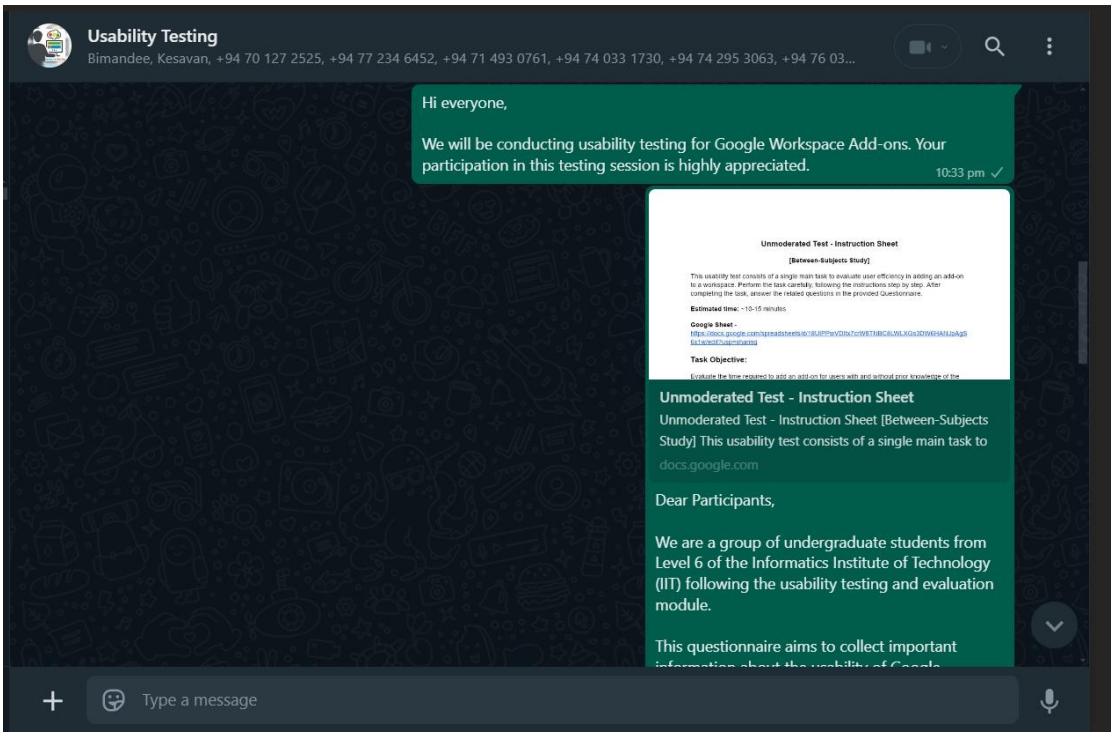



*Figure 14 Participant Recruitment Evidence 2*

# Appendix IV. Data Collection Material

IV.A Pre-Test Questionnaire: https://forms.gle/cEw5byxKZXuyy2VZ9

IV.B Post-Test Questionnaire: https://forms.gle/4GgurgX4KHJjFZXf7

IV.C Consent Form: https://forms.gle/ywvhGYCZBpDyQHt7A

## IV.A Pre-Test Questionnaire Form




# Google workspace Add-ons Usability Testing

We are a group of Undergraduate students from Level 6 of the Informatics Institute of Technology (IIT) following the **Usability testing and Evaluation** module.

The goal of this questionnaire is to collect important information about the usability of the Google Workspace Add-Ons. As part of our coursework for the BSc (Hons) Computer Science Degree at IIT, affiliated with the University of Westminster, UK. We want to enhance our understanding of user experience with Google Workspace features.

**Instructions for Participants:**

- Please take your time exploring the Google workspace Sidebar Add-ons features during the functioning evaluations.
- Share your honest opinions and observations, since your feedback is critical to our study.
- Your responds will remain confidential and used only for academic purposes.

Thank you for participating! Your valuable opinion will make a big contribution to our Usability Testing & Evaluation coursework.

**Consent to Participate**

Your privacy is of utmost importance to us. We will not be requiring you to disclose your personal information on this questionnaire. If you have disclosed any personal information during this usability test and would like it removed, we assure you that we will promptly

*Figure 15 Pre-Test Questionnaire part 1*

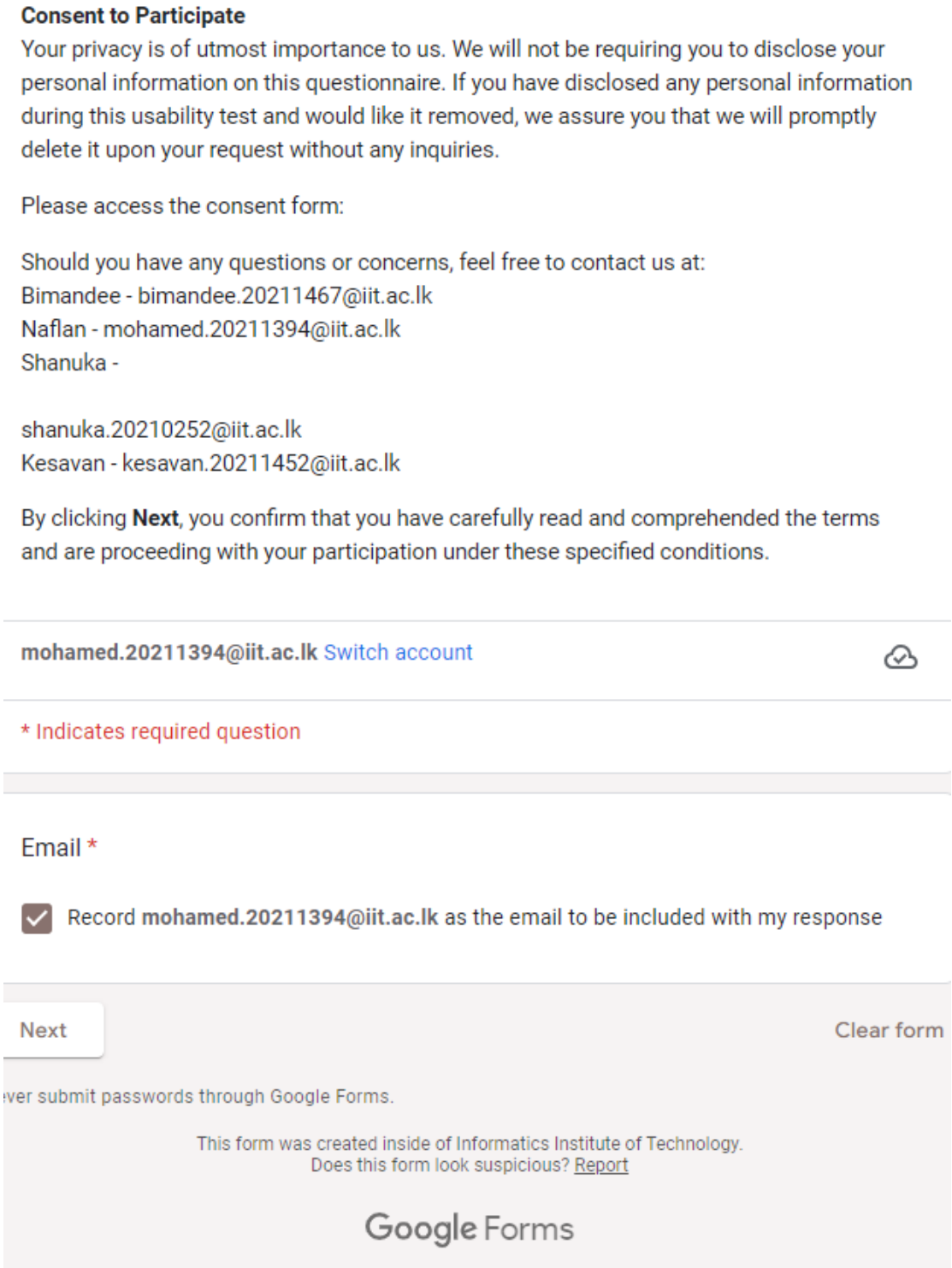



*Figure 16 Pre-Test Questionnaire part 2*

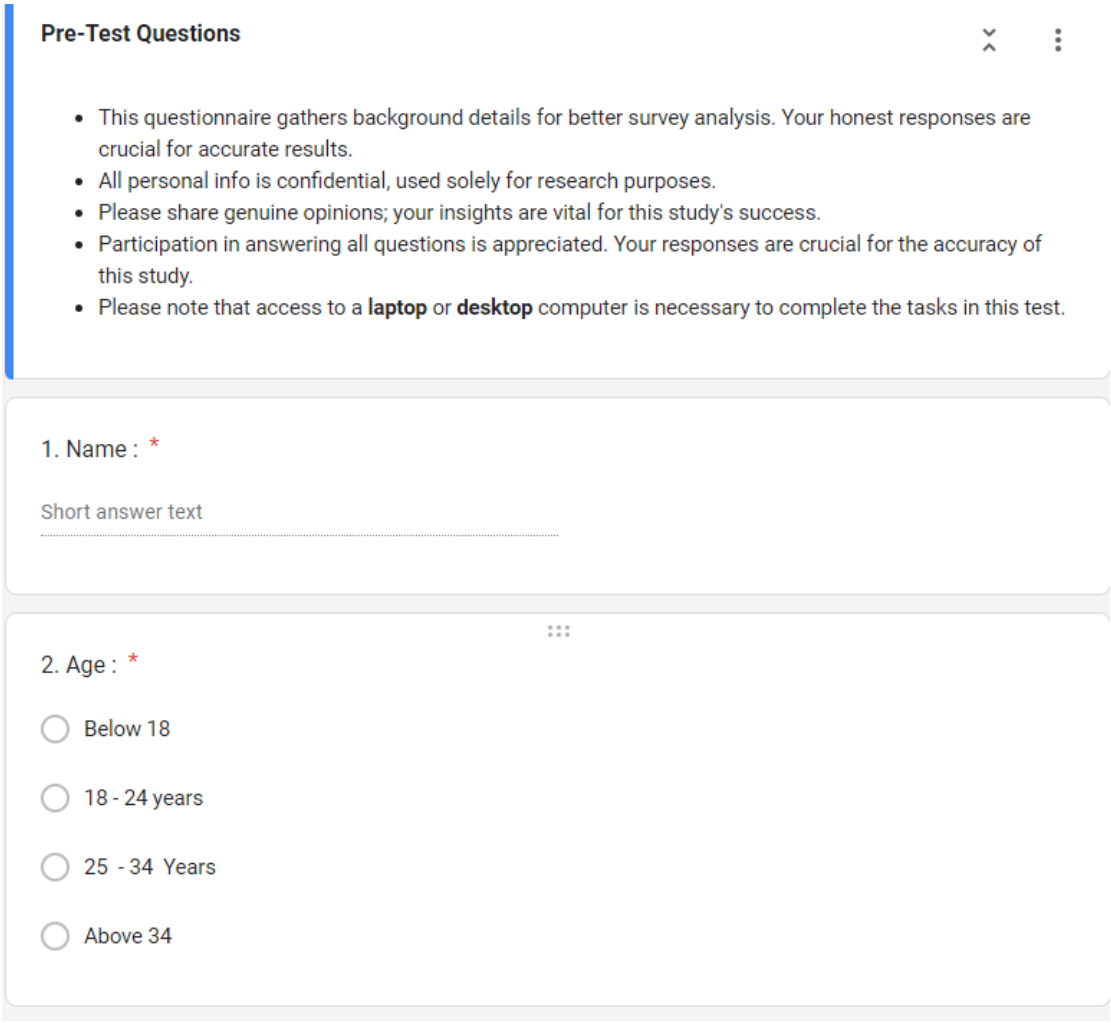



*Figure 17 Pre-Test Questionnaire part 3*

3. What is the degree program you are following: *

- Computer Science
- Software Engineering
- AI & DS
- Business Information System
- Business Management
- Business Data Analytics

4. Are you frequent user of Google Workspace Application (Google Doc, Slides, Sheets, Form)? *

- Yes
- No

5. Did you know that Google Workspace includes a sidebar located on the right side? *

- Yes
- No

*Figure 18 Pre-Test Questionnaire part 4*

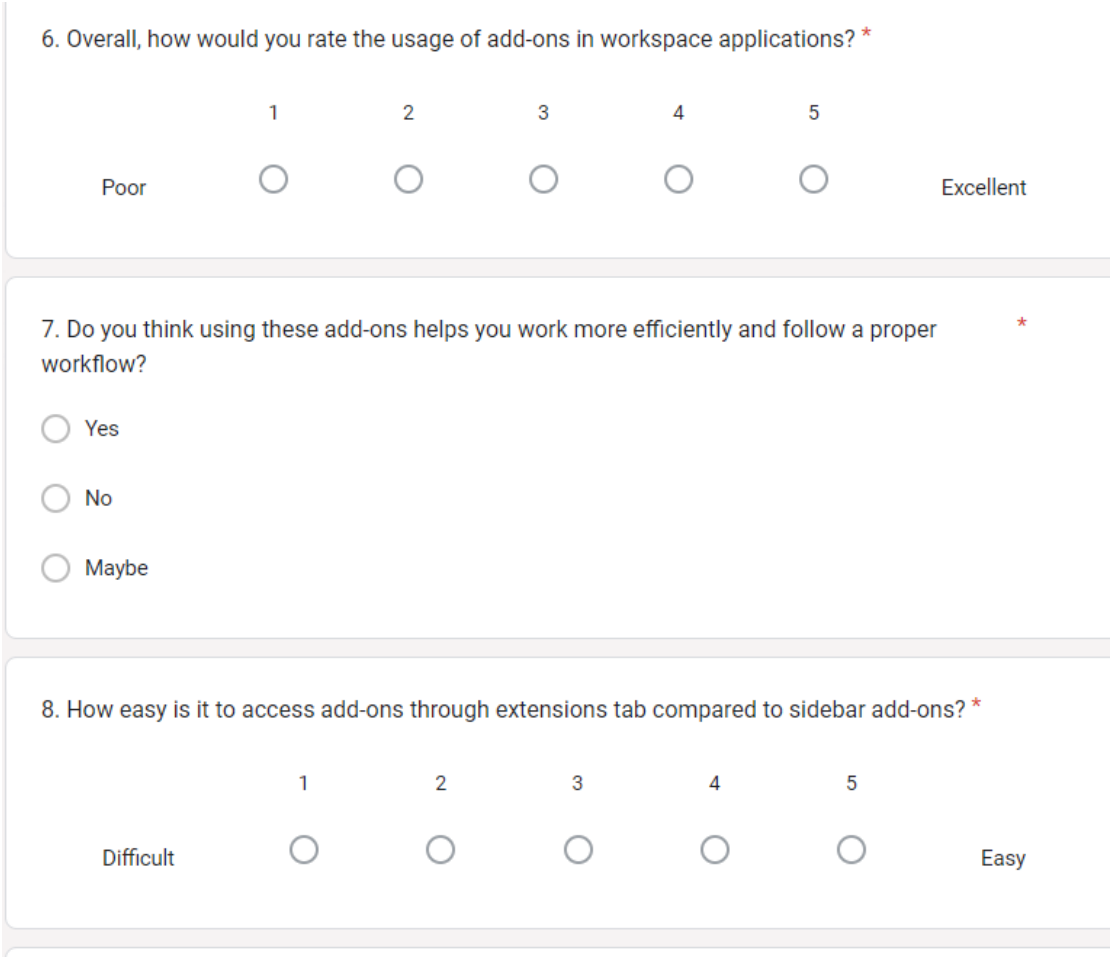



*Figure 19 Pre-Test Questionnaire part 5*

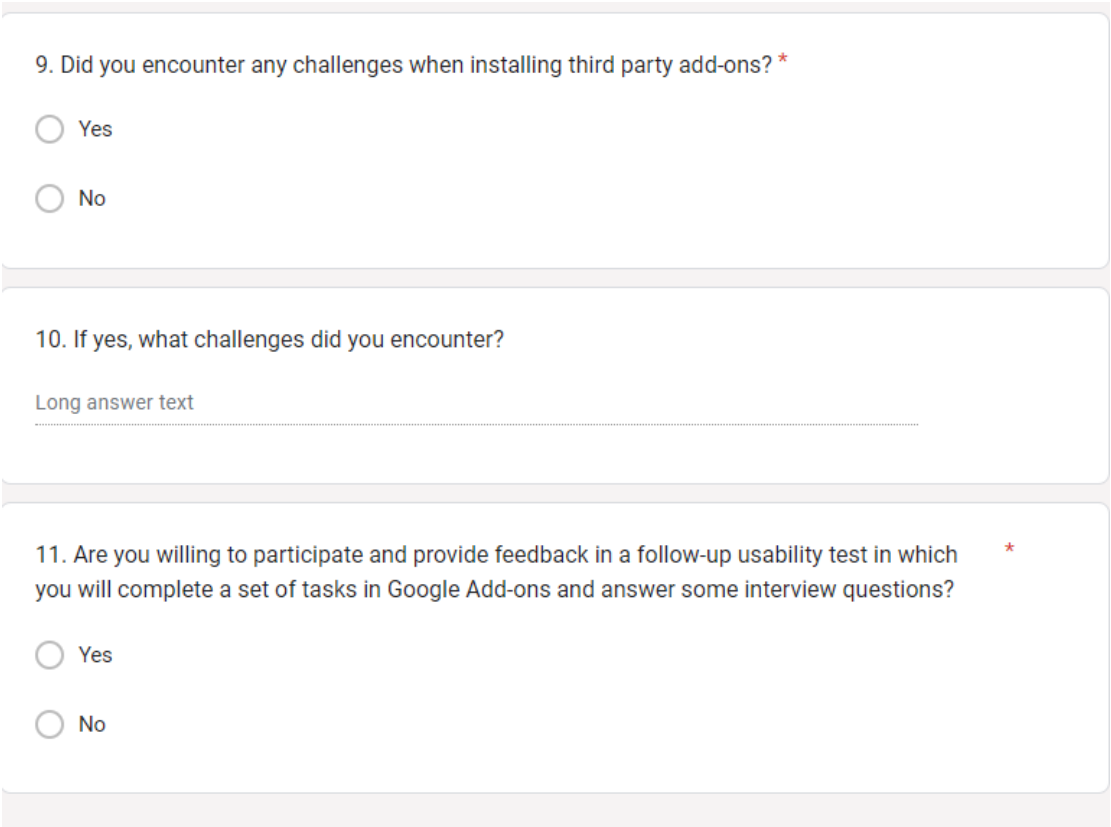
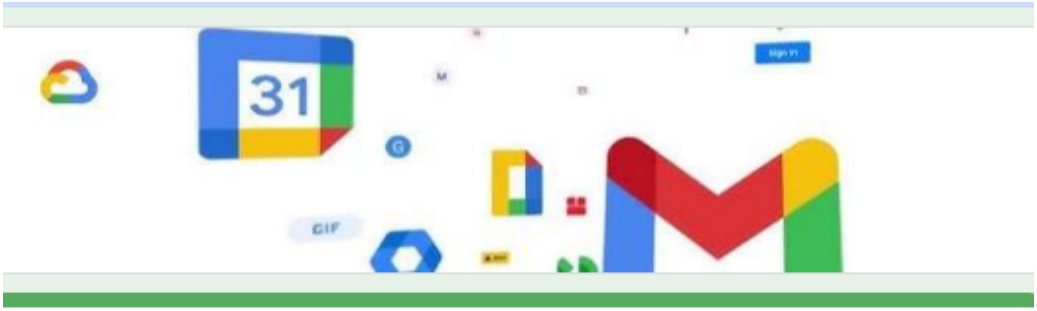



*Figure 20 Pre-Test Questionnaire part 6*

## IV.B Post-Test Questionnaire Form



# Google workspace Add-ons Usability Testing

**Post-Test Questionnaire**

We are a group of Undergraduate students from Level 6 of the Informatics Institute of Technology (IIT) following the **Usability testing and Evaluation** module.

The goal of this questionnaire is to collect important information about the usability of the Google Workspace Add-Ons. As part of our coursework for the BSc (Hons) Computer Science Degree at IIT, affiliated with the University of Westminster, UK. We want to enhance our understanding of user experience with Google Workspace features.

**Instructions for Participants:**

- Please take your time exploring the Google workspace Sidebar Add-ons features during the functioning evaluations.
- Share your honest opinions and observations, since your feedback is critical to our study.
- Your responds will remain confidential and used only for academic purposes.

Thank you for participating! Your valuable opinion will make a big contribution to our Usability Testing & Evaluation coursework.

**Consent to Participate**
Your privacy is of utmost importance to us. We will not be requiring you to disclose your personal information on this questionnaire. If you have disclosed any personal information during this usability test and would like it removed, we assure you that we will promptly delete it upon your request without any inquiries.

Please access the consent form:

Should you have any questions or concerns, feel free to contact us at:
Bimandee - bimandee.20211467@iit.ac.lk
Naflan - mohamed.20211394@iit.ac.lk
Shanuka - shanuka.20210252@iit.ac.lk
Kesavan - kesavan.20211452@iit.ac.lk

*Figure 21Post Test questionnaire*

## IVB 1. Independent Sample Post Test questionnaire.

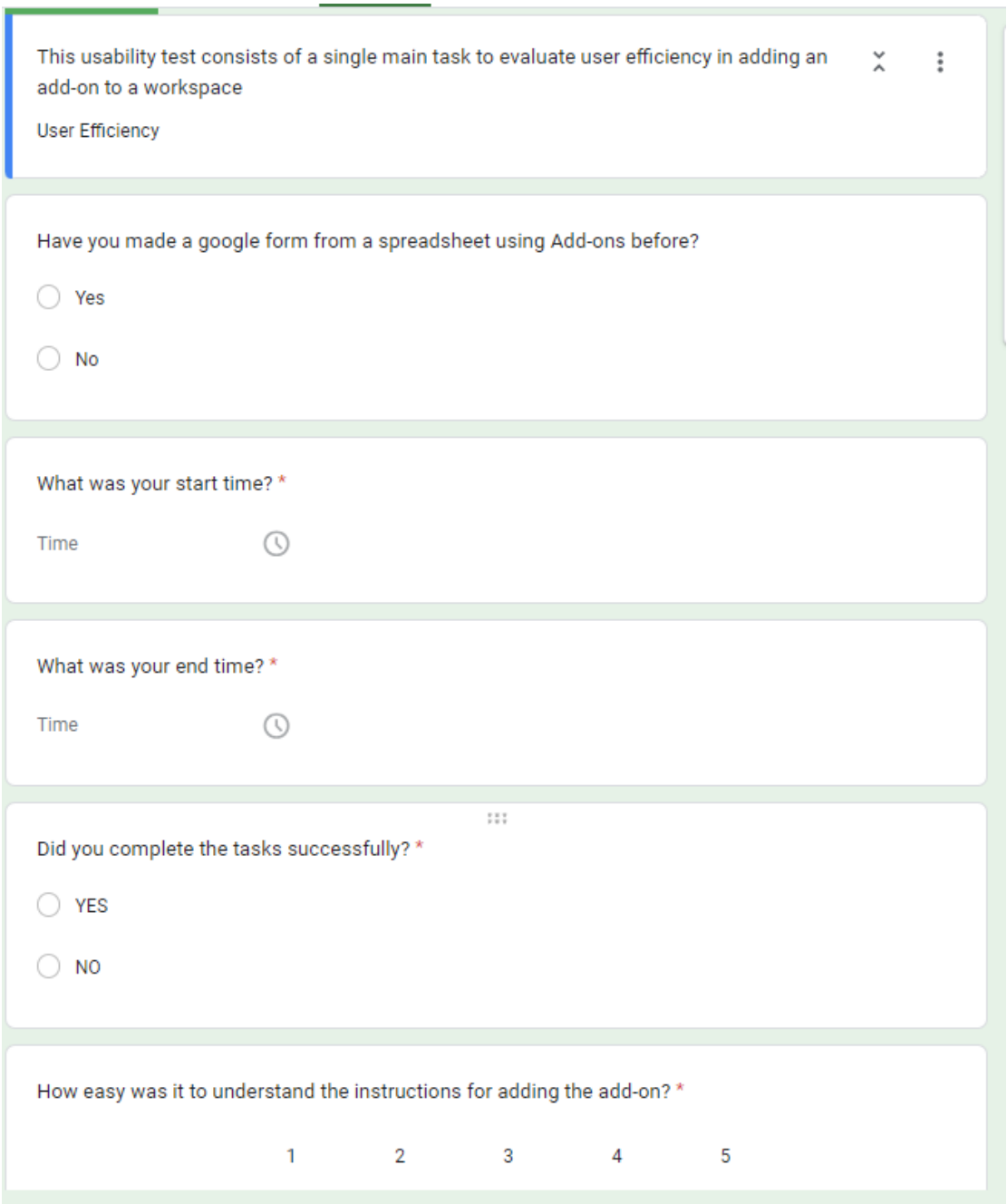
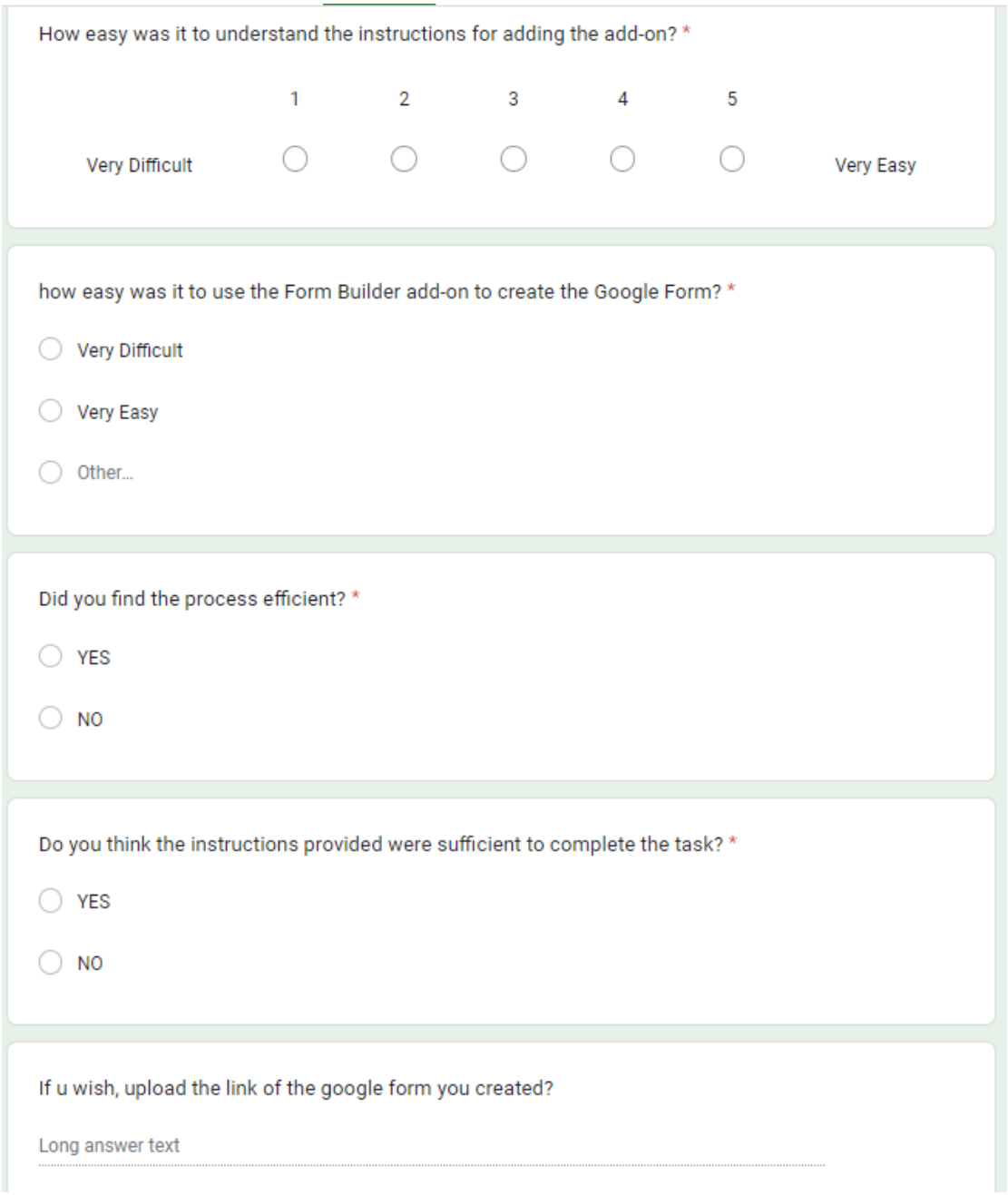



*Figure 22 Independent Sample Post Test questionnaire part 1*




***Independent Sample Post Test questionnaire part 2***

## IVB 2. Paired-Sample Post test Questionnaire

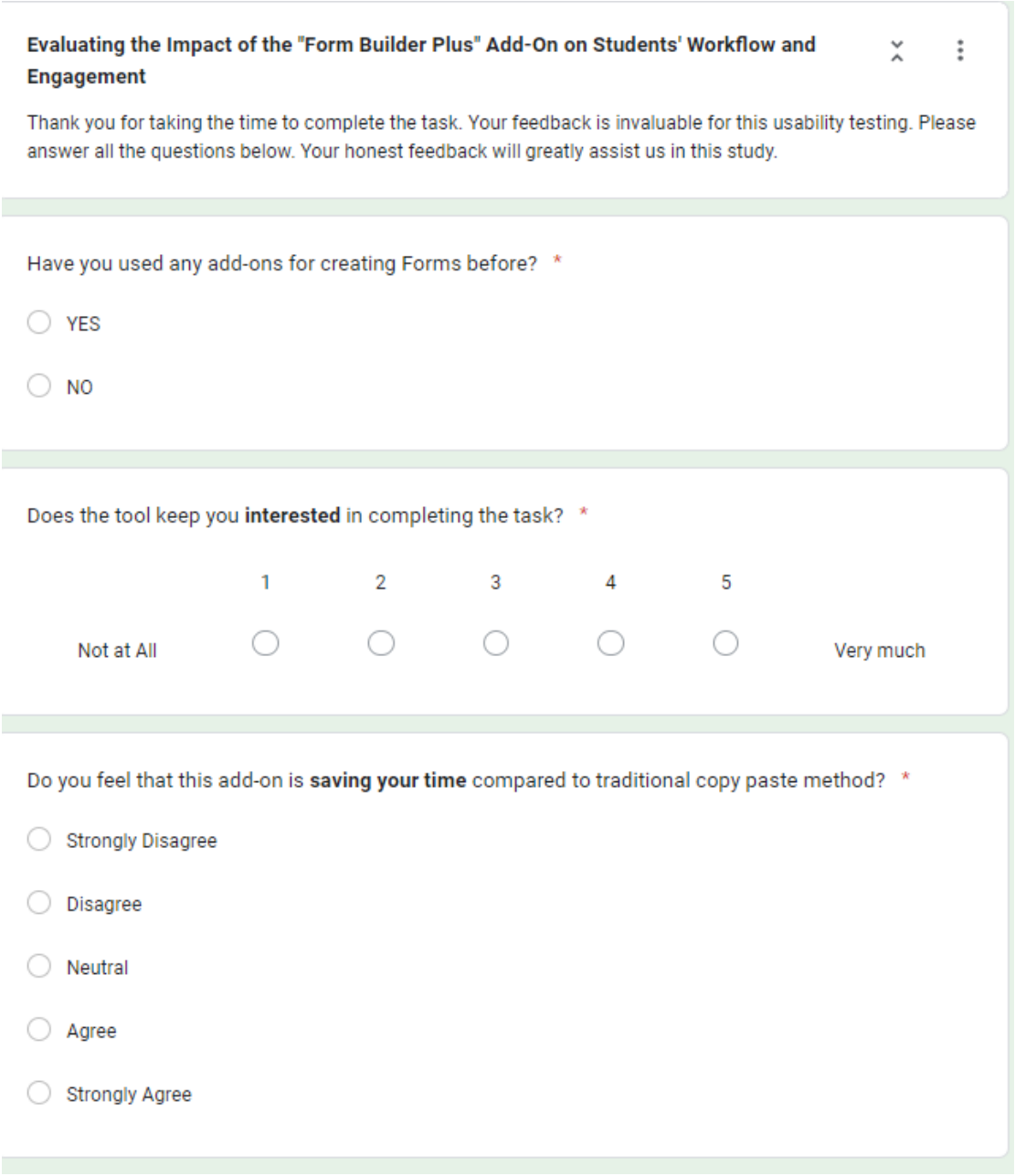



*Figure 23 Paired-Sample Post test Questionnaire Part 1*

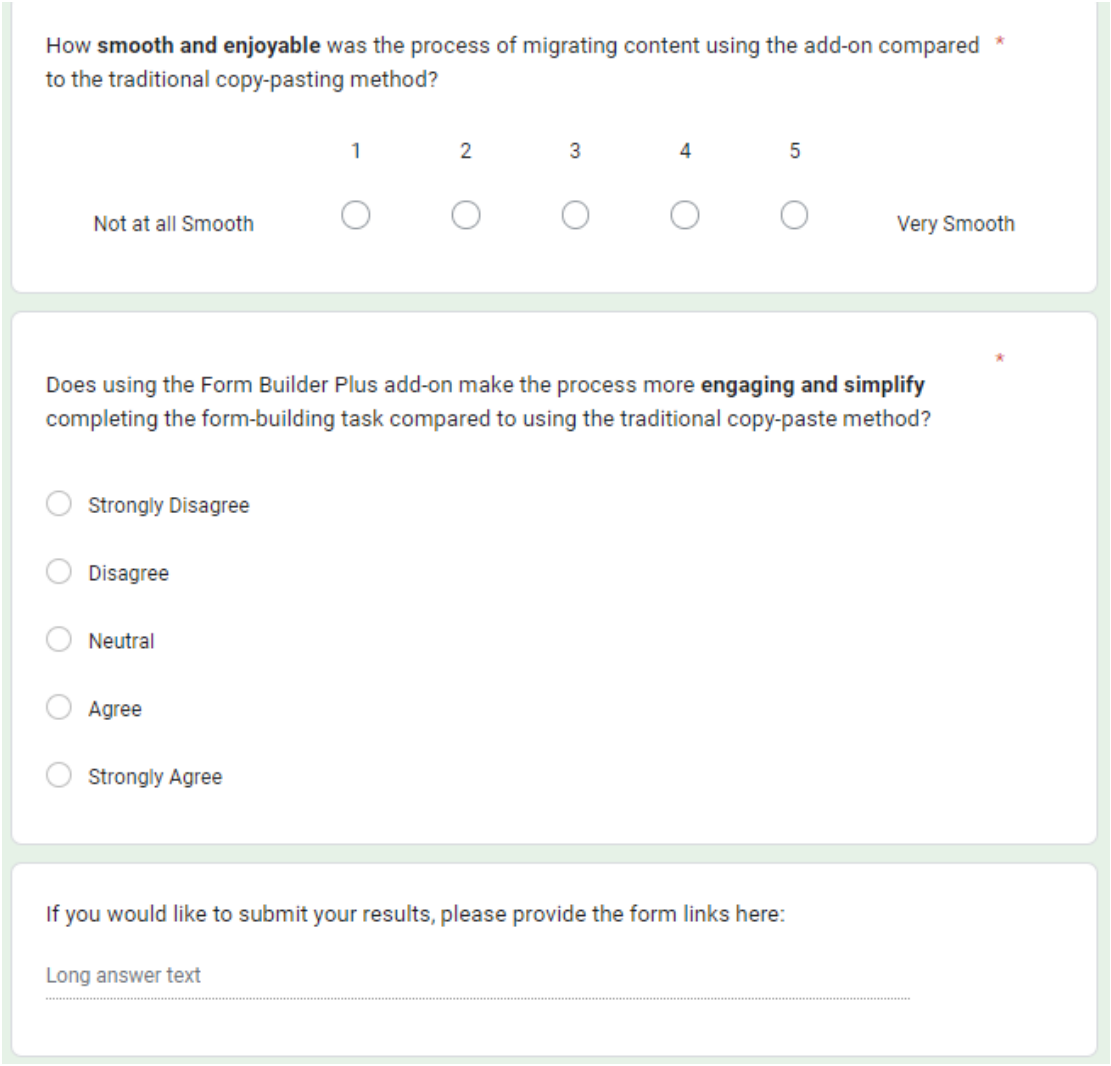



***Figure 24 Paired-Sample Post test Questionnaire Part 2***

## IVB 3. Anova Post Test Questionnaire

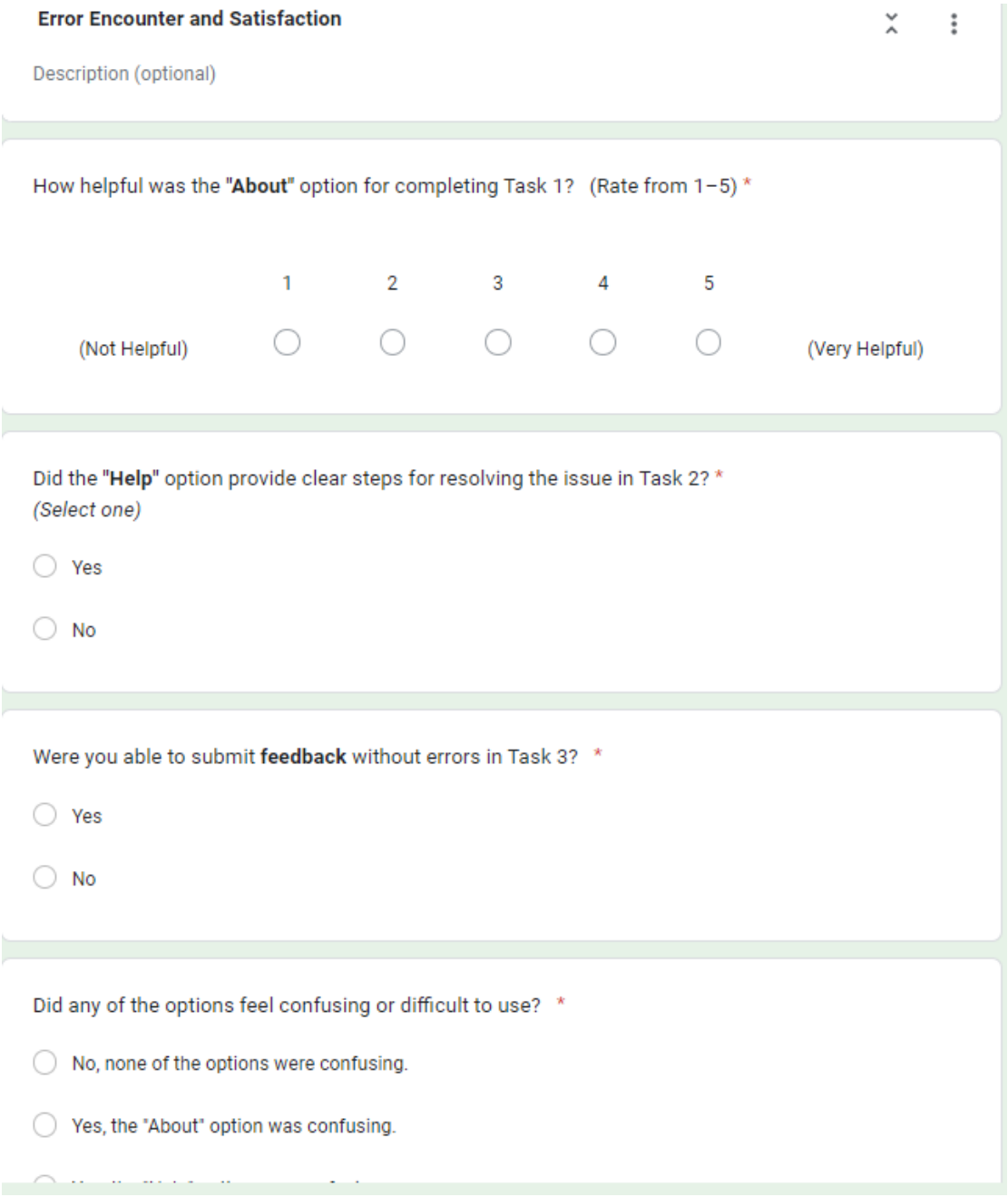



*Figure 25 Anova Post test Questionnaire part 1*

Did any of the options feel confusing or difficult to use? *

- ○ No, none of the options were confusing.
- ○ Yes, the "About" option was confusing.
- ○ Yes, the "Help" option was confusing.
- ○ Yes, the "Feedback" option was confusing.

On a scale of 1–5, how effective was the "About" option in helping you complete your task? *

| | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|
| Not Effective | ○ | ○ | ○ | ○ | ○ | Very Effective |

**On a scale of 1–5, how effective was the "Help" option in troubleshooting your problem?** *

| | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|
| Not Effective | ○ | ○ | ○ | ○ | ○ | Very Effective |

**On a scale of 1–5, how effective was the "Feedback" option for submitting your input?** *

| | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|
| Not Effective | ○ | ○ | ○ | ○ | ○ | Very Effective |

*Figure 26 Anova Post test Questionnaire Part 2*

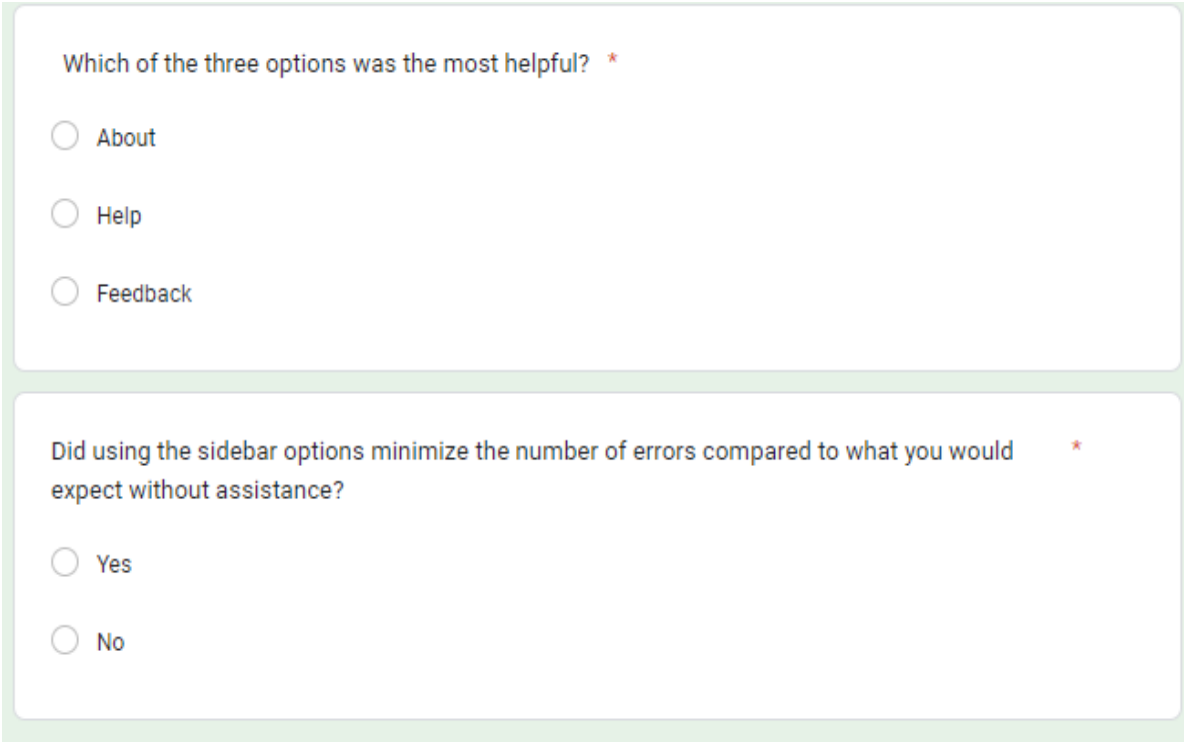
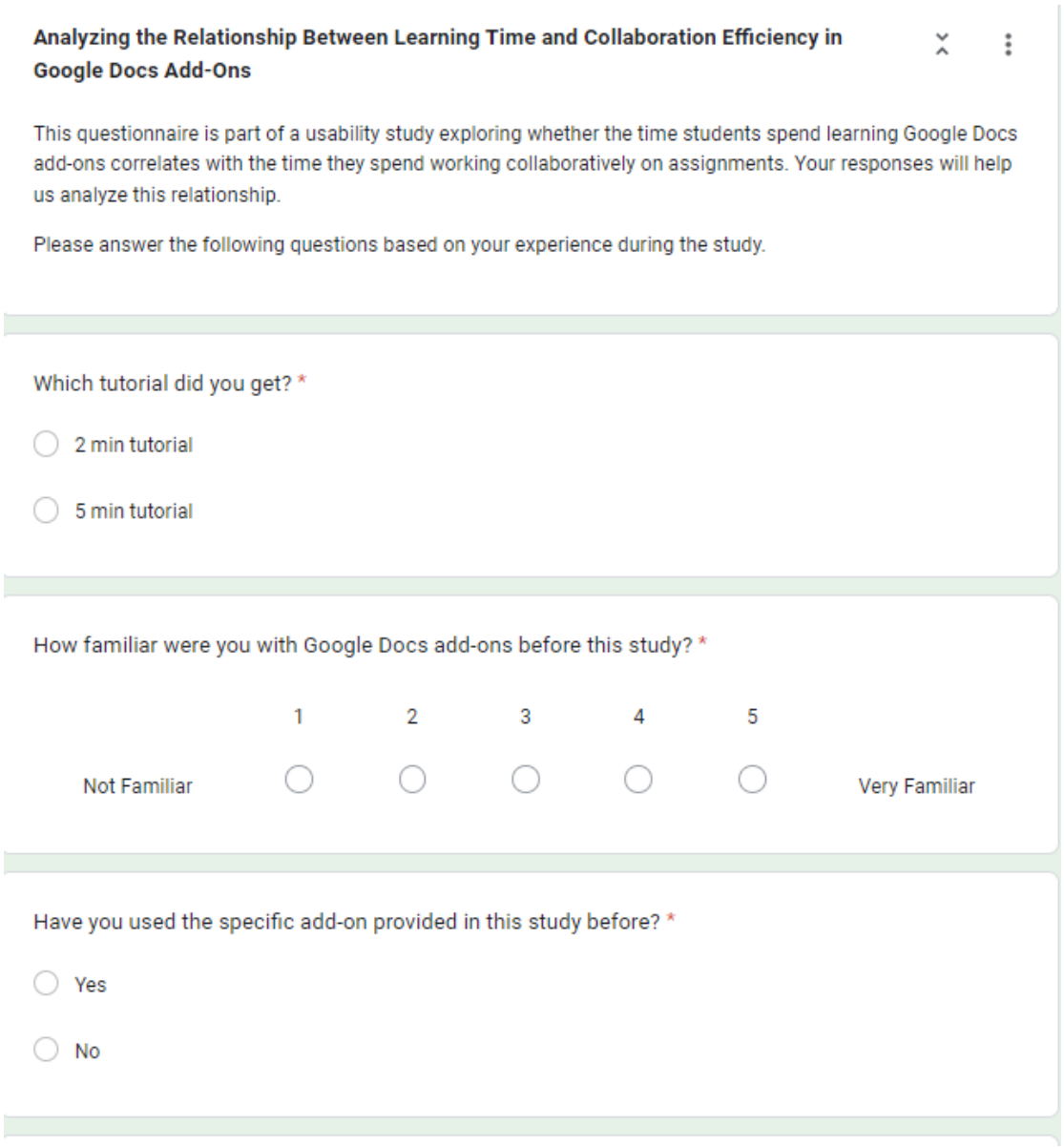



*Figure 27 Anova Post test Questionnaire part 3*

## IVB 4. Pearson Correlation Post Test Questionnaire



**Analyzing the Relationship Between Learning Time and Collaboration Efficiency in Google Docs Add-Ons**

This questionnaire is part of a usability study exploring whether the time students spend learning Google Docs add-ons correlates with the time they spend working collaboratively on assignments. Your responses will help us analyze this relationship.

Please answer the following questions based on your experience during the study.

Which tutorial did you get? *

- 2 min tutorial
- 5 min tutorial

How familiar were you with Google Docs add-ons before this study? *

| | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|
| Not Familiar | | | | | | Very Familiar |

Have you used the specific add-on provided in this study before? *

- Yes
- No

*Figure 28 Pearson Correlation Post Test Questionnaire part 1*

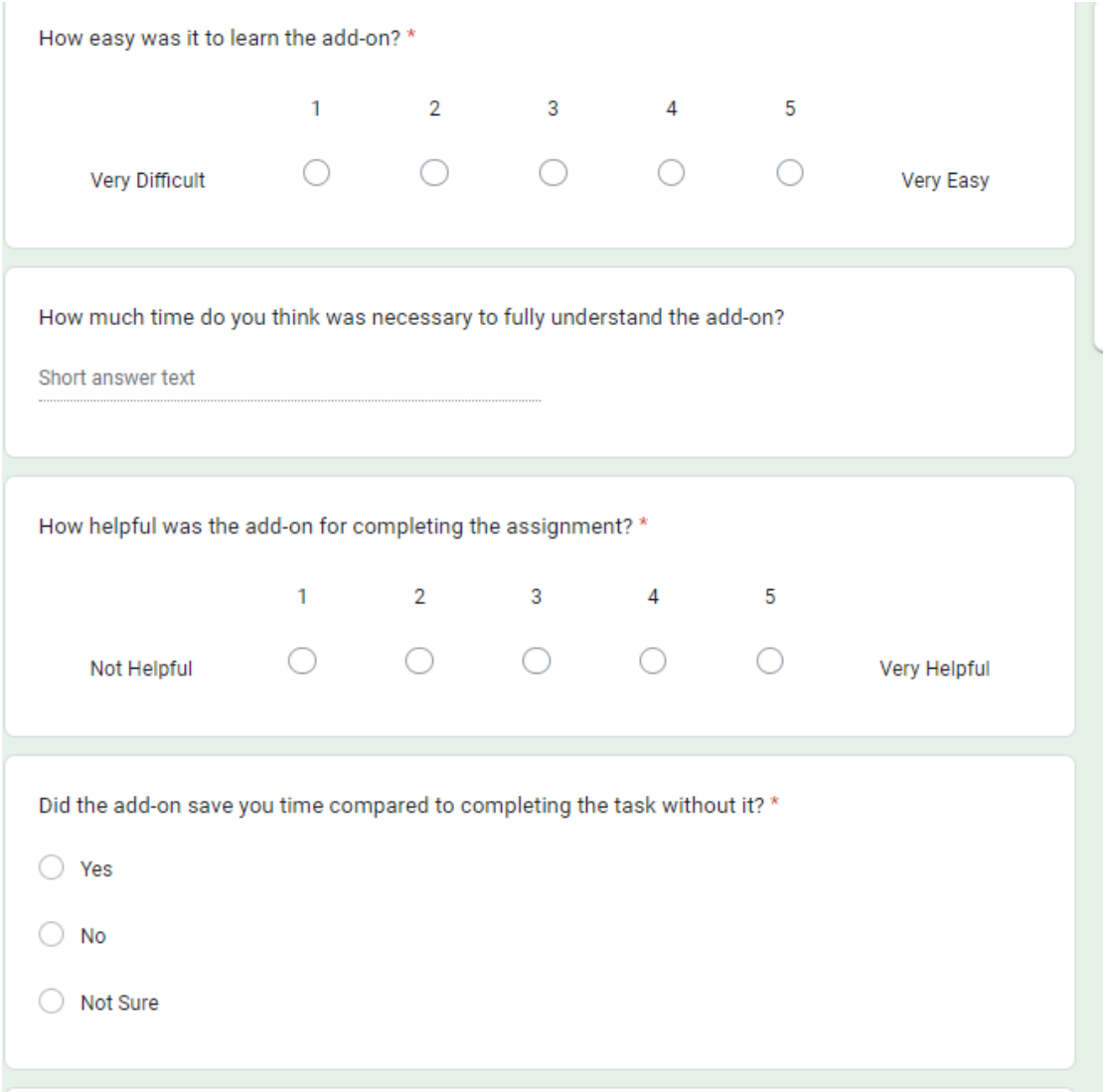



***Figure 29 Pearson Correlation Post Test Questionnaire part 2***

## IVC Consent Form

# CONSENT FORM

**PARTICIPATION INFORMATION**

**Usability testing and Evaluation Google Workspace Add- ons**

**Researcher(s):** Kesavan Anbuchelvan Mohamed Naflan Bimandee Premarathna Shanuka Karunarathne

**Supervisors:** Nishantha Janith Chandrasena

You are invited to participate in research study on learning about a set of add-ons integrated into the sidebar of Google Workspace, which includes a series of tasks and surveys.This study was undertaken to provide students at the university with a solid knowledge of Google add-ons incorporated into the sidebar, in order to make their work easier when working in Google workspace. The goal of this Usability study and testing is to collect the data required to evaluate the usability of different add-ons for facilitating collaboration, document management, and task organization. Our research questions and hypotheses will center on how these tools can improve students' collaborative and productive experiences in Google Workspace.

This study is being conducted as part of the researcher's coursework for the university's Usability Testing Module in Computer Science.

The study will involve you:

1. Completing a specified set of tasks pertaining to Google Workspace's sidebar add-ons that are intended to collect information needed for the usability test. This should take about 60 minutes .
2. Participants should follow the task instruction page and fill out three questionnaires at the same time. The purpose of these forms is to collect information about the use of add-ons, user happiness, task completion times, and feedback on their use.

*Figure 30 Consent form part 1*

**Participation in Research**

- Your participation in this research is entirely voluntary. You have the right to withdraw at any time without giving a reason.
- Withdrawal from the research will not affect any treatment and/or services that you receive.
- You can request your data to be withdrawn and personal information destroyed, as long as it is practical.
- You are not required to answer particular questions in questionnaires or interviews if you do not wish to.

**Privacy and Confidentiality**

- Your responses will normally be made anonymous and kept confidential unless you provide explicit consent to do otherwise, such as for the use of your image in photographs or video recordings.
- No individuals will be identifiable from collated data, written reports, or any publications arising from this research.

**Data Security**

- All computer data files will be encrypted and password protected. The researcher will store files securely and comply with the requirements of the Data Protection Act.
- Hard copy documents, such as consent forms and completed questionnaires, will be kept securely in a locked cupboard, preferably on university premises.
- Documents may be scanned and stored electronically to enable secure transmission to the university's secure computer systems.

**Post-Research Communication**

- If you wish, you can request information on the research results by indicating this on the consent form.
- The researcher can be contacted via email after your participation.

*Figure 31Consent form part 2*

**Post-Research Communication**

- If you wish, you can request information on the research results by indicating this on the consent form.
- The researcher can be contacted via email after your participation.

**Complaints**

- If you have a complaint about this research project, you can contact the project supervisor, Nishantha Janith Chandrasena via an email.

mohamed.20211394@iit.ac.lk Switch account

* Indicates required question

Email *

Record mohamed.20211394@iit.ac.lk as the email to be included with my response

Next Clear form

*Figure 32 Consent form part 3*

**PARTICIPATION INFORMATION - CONSENT FORM**

Description (optional)

Participant Name: *

Short answer text

Date : *

Month, day, year

I have been given the Participation Information Sheet and/or had its contents explained to me. *

YES

NO

I have had an opportunity to ask any questions and I am satisfied with the answers given. *

YES

*Figure 33 Consent form part 4*

I understand I have a right to withdraw from the research at any time and I do not have to provide a reason. *

- YES
- NO

I understand that if I withdraw from the research any data included in the results will be removed if that is practicable (I understand that once anonymised data has been collated into other datasets it may not be possible to remove that data). *

- YES
- NO

I would like to receive information relating to the results from this study. *

- YES
- NO

*Figure 34 Consent form part 5*

I wish to receive a copy of this Consent form. *

- YES
- NO

I confirm I am willing to be a participant in the above research study. *

- YES
- NO

I note the data collected may be retained in an archive and I am happy for my data to be reused as part of future research activities. I note my data will be fully anonymised. *

- YES
- NO

*Figure 35 Consent form part 6*

# Appendix V. Test Recordings

## VA. Usability testing session photos



*Figure 36 Usability Testing Session picture1*



*Figure 37 Usability Testing Session picture 2*

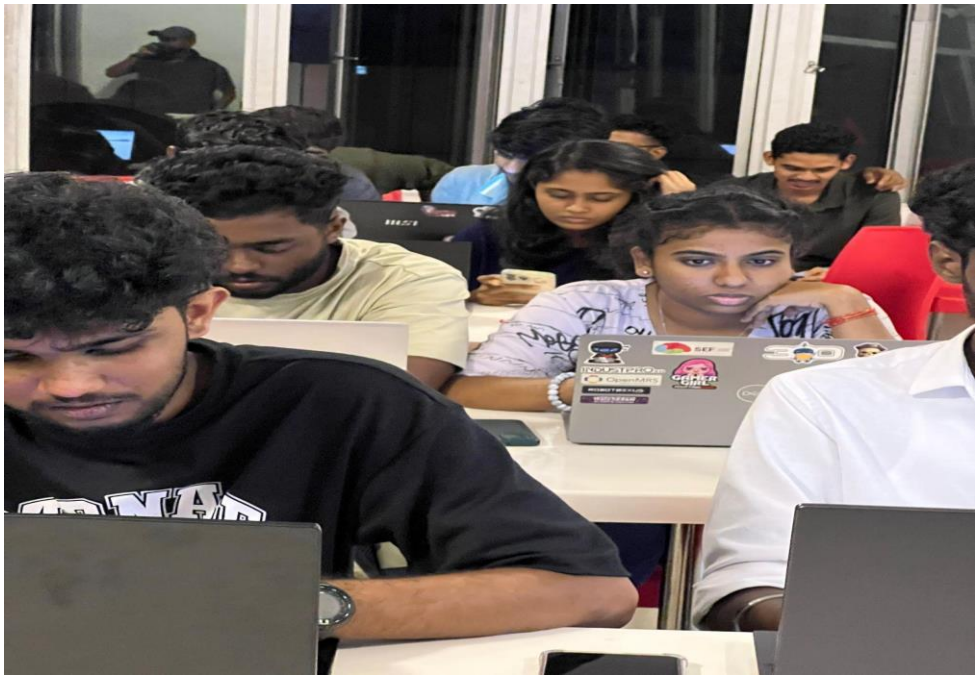


*Figure 38 Usability Testing Session picture3*



*Figure 39 Usability Testing Session picture4*



*Figure 40 Usability Testing Session picture 5*



*Figure 41Usability Testing Session picture 6*

**VB. Usability testing session Videos Link**

https://youtube.com/shorts/ci9nUZz_pQI?si=IJQTuJXa3fsd_Cit

https://youtube.com/shorts/MWfPkOSAjp8?si=-LRnewLB4Pzm5vOU

https://youtu.be/Tttv5pgmo7M?si=QF6VmD4cM6MDGTQB

https://youtu.be/km2LV3QItHM?si=9Jsxzdq0DNW0jD0h

https://youtube.com/shorts/z7JqRHbPejw?si=tUnReStEIn3imULs

https://youtube.com/shorts/uCwfrl0y4_c?si=37Cf3ZbMgOEgS8DX

https://youtu.be/rcfS8CaSEuA?si=Rc6kyQN8m2NERTeU

https://youtu.be/TD8QD-jc1og?si=dziO11-YyhHzxgPd

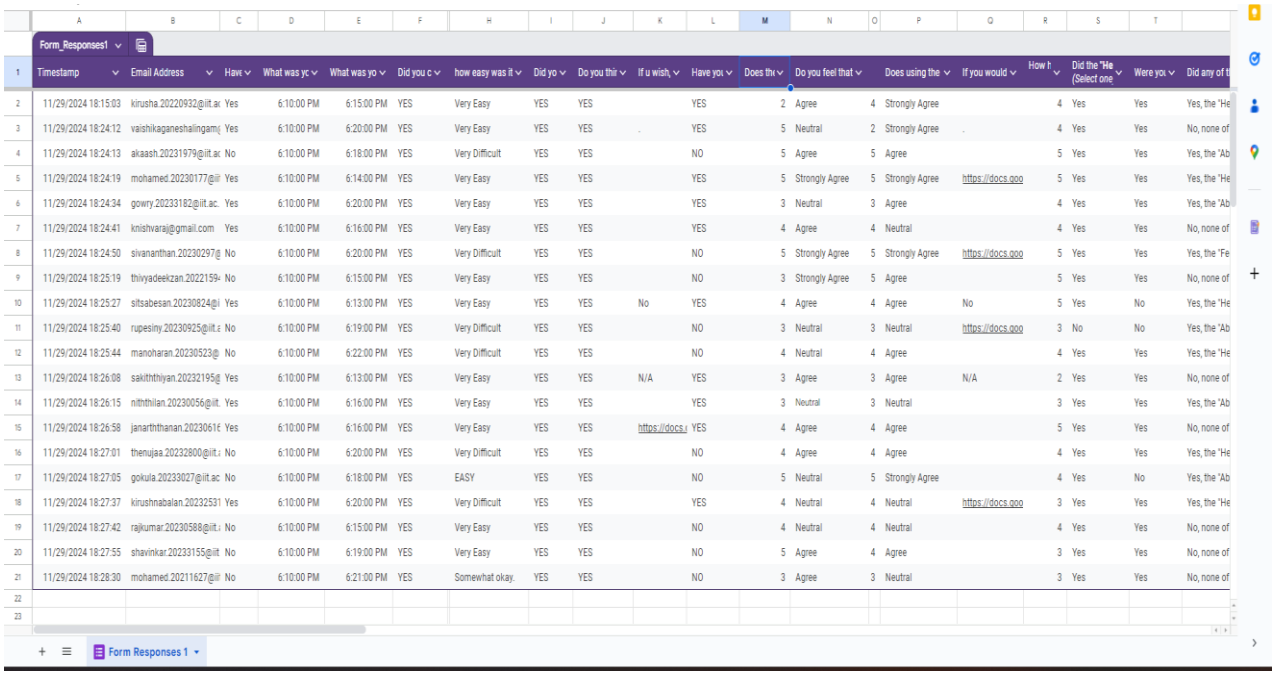
https://youtu.be/iok7oDKc394?si=Rcp7zNG37cWt_UX8

https://youtu.be/N-q-h9ut8WY?si=e3yXfeFM2ToeRapF

https://www.youtube.com/watch?v=LE97o7tukqg

## Appendix VI. Raw data

Here we have attached links to the Google Form that collects the results from the Google Questionnaire.



| Timestamp | Email Address | Have | What was yc | What was yo | Did you c | how easy was it | Did yo | Do you thir | If u wish, | Have you | Does thi | Do you feel that | | Does using the | | If you would | How h | Did the "He (Select one | Were you | Did any of t |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 11/29/2024 18:15:03 | kirusha.20220932@iit.ac | Yes | 6:10:00 PM | 6:15:00 PM | YES | Very Easy | YES | YES | | YES | 2 | Agree | 4 | Strongly Agree | | | 4 | Yes | Yes | Yes, the "He |
| 11/29/2024 18:24:12 | vaishikaganeshalingam( | Yes | 6:10:00 PM | 6:20:00 PM | YES | Very Easy | YES | YES | . | YES | 5 | Neutral | 2 | Strongly Agree | . | | 4 | Yes | Yes | No, none of |
| 11/29/2024 18:24:13 | akaash.20231979@iit.ac | No | 6:10:00 PM | 6:18:00 PM | YES | Very Difficult | YES | YES | | NO | 5 | Agree | 5 | Agree | | | 5 | Yes | Yes | Yes, the "Ab |
| 11/29/2024 18:24:19 | mohamed.20230177@iit | Yes | 6:10:00 PM | 6:14:00 PM | YES | Very Easy | YES | YES | | YES | 5 | Strongly Agree | 5 | Strongly Agree | https://docs.goo | | 5 | Yes | Yes | Yes, the "He |
| 11/29/2024 18:24:34 | gowry.20233182@iit.ac. | Yes | 6:10:00 PM | 6:20:00 PM | YES | Very Easy | YES | YES | | YES | 3 | Neutral | 3 | Agree | | | 4 | Yes | Yes | Yes, the "Ab |
| 11/29/2024 18:24:41 | knishvaraj@gmail.com | Yes | 6:10:00 PM | 6:16:00 PM | YES | Very Easy | YES | YES | | YES | 4 | Agree | 4 | Neutral | | | 4 | Yes | Yes | No, none of |
| 11/29/2024 18:24:50 | sivananthan.20230297@ | No | 6:10:00 PM | 6:20:00 PM | YES | Very Difficult | YES | YES | | NO | 5 | Strongly Agree | 5 | Strongly Agree | https://docs.goo | | 5 | Yes | Yes | Yes, the "Fe |
| 11/29/2024 18:25:19 | thivyadeekzan.2022159 | No | 6:10:00 PM | 6:15:00 PM | YES | Very Easy | YES | YES | | NO | 3 | Strongly Agree | 5 | Agree | | | 5 | Yes | Yes | No, none of |
| 11/29/2024 18:25:27 | sitsabesan.20230824@i | Yes | 6:10:00 PM | 6:13:00 PM | YES | Very Easy | YES | YES | No | YES | 4 | Agree | 4 | Agree | No | | 5 | Yes | No | Yes, the "He |
| 11/29/2024 18:25:40 | rupesiny.20230925@iit.a | No | 6:10:00 PM | 6:19:00 PM | YES | Very Difficult | YES | YES | | NO | 3 | Neutral | 3 | Neutral | https://docs.goo | | 3 | No | No | Yes, the "Ab |
| 11/29/2024 18:25:44 | manoharan.20230523@ | No | 6:10:00 PM | 6:22:00 PM | YES | Very Difficult | YES | YES | | NO | 4 | Neutral | 4 | Agree | | | 4 | Yes | Yes | Yes, the "He |
| 11/29/2024 18:26:08 | sakiththiyan.20232195@ | Yes | 6:10:00 PM | 6:13:00 PM | YES | Very Easy | YES | YES | N/A | YES | 3 | Agree | 3 | Agree | N/A | | 2 | Yes | Yes | No, none of |
| 11/29/2024 18:26:15 | niththilan.20230056@iit. | Yes | 6:10:00 PM | 6:16:00 PM | YES | Very Easy | YES | YES | | YES | 3 | Neutral | 3 | Neutral | | | 3 | Yes | Yes | Yes, the "Ab |
| 11/29/2024 18:26:58 | janarththanan.2023061( | Yes | 6:10:00 PM | 6:16:00 PM | YES | Very Easy | YES | YES | https://docs.( | YES | 4 | Agree | 4 | Agree | | | 5 | Yes | Yes | No, none of |
| 11/29/2024 18:27:01 | thenujaa.20232800@iit.( | No | 6:10:00 PM | 6:20:00 PM | YES | Very Difficult | YES | YES | | NO | 4 | Agree | 4 | Agree | | | 4 | Yes | Yes | Yes, the "He |
| 11/29/2024 18:27:05 | gokula.20233027@iit.ac | No | 6:10:00 PM | 6:18:00 PM | YES | EASY | YES | YES | | NO | 5 | Neutral | 5 | Strongly Agree | | | 4 | Yes | No | Yes, the "Ab |
| 11/29/2024 18:27:37 | kirushnabalan.20232531 | Yes | 6:10:00 PM | 6:20:00 PM | YES | Very Difficult | YES | YES | | YES | 4 | Neutral | 4 | Neutral | https://docs.goo | | 3 | Yes | Yes | Yes, the "He |
| 11/29/2024 18:27:42 | rajkumar.20230588@iit.( | No | 6:10:00 PM | 6:15:00 PM | YES | Very Easy | YES | YES | | NO | 4 | Neutral | 4 | Neutral | | | 4 | Yes | Yes | No, none of |
| 11/29/2024 18:27:55 | shavinkar.20233155@iit | No | 6:10:00 PM | 6:19:00 PM | YES | Very Easy | YES | YES | | NO | 5 | Agree | 4 | Agree | | | 3 | Yes | Yes | No, none of |
| 11/29/2024 18:28:30 | mohamed.20211627@iit | No | 6:10:00 PM | 6:21:00 PM | YES | Somewhat okay. | YES | YES | | NO | 3 | Agree | 3 | Neutral | | | 3 | Yes | Yes | No, none of |

*Figure 42 Usability Testing Results Sample*

**Links for the testing results.**

Pre-Test Questionnaire: Pre-Test Questionnaire (Responses)

Post-Test Questionnaire: Post-Test Questionnaire (Responses)

Consent Form: Consent Form - Google Workspace Add-ons (Responses)

Usability Testing results Summary: UT Result Summary

## Appendix VII. Ethics form

Part A Research-Ethics Consideration Application Form: Part-A-Research-Ethics-Consideration

Ethics form Cover Sheet: Cover-sheet2020_21.doc

---

UNIVERSITY OF WESTMINSTER

**University Research Ethics Committee**

**Front cover sheet for applications**

**Applicant's name:** Kesavan, Naflan, Shanuka, Bimandee

**Project Title: Usability Testing – Google Workspace Add-ons**

**Application Reference** ___ - ___ - ___ **(for office use only)**

**Please complete the checklist below before submitting your ethics application**

| **Enclosed:** | **YES** | **NO** | **N/A** |
|---|---|---|---|
| Application Form Part A attached. | ☒ | ☐ | ☐ |
| Application Form Part B attached (if applicable). | ☐ | ☐ | ☒ |
| Any external ethical approval (copy of application and approval letter) attached. | ☐ | ☐ | ☒ |
| Draft Participant Information Sheet attached (see available exemplars). | ☒ | ☐ | ☐ |
| Draft Informed Consent Form attached (see available exemplars). | ☒ | ☐ | ☐ |
| Draft Indicative Questions, e.g. questionnaire(s), proposed interview questions or questioning areas, etc. attached. | ☒ | ☐ | ☐ |
| Appropriate risk assessments have been completed, e.g. Control of Substances Hazardous to Health (COSHH), Radiation, etc. (if applicable) – *Contact the University's Safety, Health and Wellbeing Team for advice on this and other aspects of health and safety.* | ☐ | ☐ | ☒ |
| Fieldwork Risk Assessment attached (if applicable). *(UCEA Guidance on Health and Safety in Fieldwork Including offsite visits and travel in the UK and overseas) – Contact the University's Safety, Health and Wellbeing Team for advice on this and other aspects of health and safety.* | ☐ | ☐ | ☒ |
| Travel Insurance Request clearance notification attached (if applicable). *Contact - Andrew Clarke (a.clarke03@westminster.ac.uk) or Alison Sylvestre (a.sylvestre@westminster.ac.uk) in Procurement if advice is required – This is essential if there is any Foreign and Commonwealth Office or RED24 advice against travel.* | ☐ | ☐ | ☐ |
| Confirmation of Insurance coverage for research undertaken off campus. | ☐ | ☐ | ☒ |
| Security-sensitive research assessment completed (if applicable) and uploaded (see UniversitiesUK Guidance and, if applicable, complete the Annex to Part B and upload). | ☐ | ☐ | ☒ |
| Other (please specify, e.g. letters from collaborators, etc.): | ☐ | ☐ | ☒ |

_____

**Applicant's signature:** Kesavan, Naflan, Shanuka, Bimandee **Date:** 27/11/2024

**Supervisor's or Faculty Research Director's signature:** _____ **Date:** _____

**PLEASE RETURN THIS COVER SHEET ALONG WITH THE REQUIRED DOCUMENTS BY EMAIL ATTACHMENT TO:**
**SECRETARY, FACULTY RESEARCH ETHICS COMMITTEE (or Contact your Supervisor).**

V1.3 9 October 2013 (updated for contact details: 19 Aug 2016) Academic Registrar's Dept.

---

*Figure 43 Consent Form cover page*

UNIVERSITY OF WESTMINSTER

OFFICE USE: ___ - ___-____

**University of Westminster**
**University Research Ethics Committee**

**Application for Research Ethics**

## PART A

| **Section 1 – PROJECT AND APPLICANT DETAILS** |
|---|

| 1. **Project Title:** **Usability Testing & Evaluation - Group Coursework (Google Workspace Add ons)** |
|---|

| **1.1 Applicant Details** | |
|---|---|
| **Name:**<br>Kesavan<br>Naflan<br>Shanuka<br>Bimandee | **University Email Address:**<br>kesavan.20211452@iit.ac.lk<br>mohamed.20211394@iit.ac.lk<br>shanuka.20210252@iit.ac.lk<br>bimandee.20211467@iit.ac.lk |
| **Contact Address:**<br>IIT Sri Lanka | **Telephone Number:**<br>+94 76 4532238<br>+94 76 732 1365<br>+94 70 127 2525<br>+94 77 669 2101 |
| **Faculty: Computer Faculty** | |
| **Please check the relevant box:**<br><br>**Undergraduate** ☒ **Postgraduate** ☐ **MPhil/PhD Student** ☐ **Staff** ☐ | |
| **I confirm I have read the *University's Code of Practice Governing the Ethical Conduct of Research*** | **YES** ☒ **NO** ☐ |

**1.3 Supervisor/Dean of Faculty/Faculty Research Director details**

Please note that all applicants with a supervisor(s) must ensure that the supervisor signs the declaration at the bottom of this page if completing Part A only or in **Section 10.3** if completing Part B

All **staff** must ensure that their Dean of Faculty, or Faculty Research Director (or nominee), as appropriate, signs the declaration at the bottom of this page if completing Part A only or in **Section 10.3** if completing Part B

| **Name:**<br>Nishantha Janith Chandrasena | **University Email Address:**<br>nishantha.w@iit.ac.lk |
|---|---|
| **Faculty:**<br>Computer Science and Engineering | **Telephone Number:** |

*Figure 44 Research Ethics Part A - Application - Part -1*

## PART A (Continued)

**Section 2 – Project Details**

**2.1** Please provide a description of the background with references to relevant literature (250 words maximum):
Usability studies are crucial for evaluating whether a product meets the needs and expectations of its users. The goal of this research is to conduct a usability test for "Google Workspace Add-ons" a collection of tools designed for Google Workspace users. These add-ons aim to enhance productivity within the Google Workspace environment, allowing users to engage more effectively with their daily tasks in G Suite applications. This study seeks to offer valuable insights to improve the usability and functionality of Google Workspace add-ons, ensuring an optimal experience for end users.

**2.2.** Please provide a brief description and the aims of your study (250 words maximum):

This research study aims to assess the usability of the 'Google Workspace Add-ons service through a focused usability test with students. The objective is to collect key data to evaluate specific aspects of the add-ons based on established usability metrics. The gathered data will be securely stored and analyzed to address the research questions and hypotheses, identifying areas for improvement and necessary adjustments. Ultimately, the study seeks to ensure that Google Workspace add-ons align with the expectations and needs of end-users.

**2.3.** Please outline the design and methodology of your study (include details of the selection and recruitment of participants (if any) and details of any invasive (e.g. blood samples, inhalation/ingestion of food and/or non-food products (in abnormally higher or lower levels than normal or a different form), or intrusive (e.g. questionnaires, focus groups, interviews, etc.) procedures [attach extra information as necessary] (400 words maximum in total):

This research will be based on hypotheses related to specific aspects of Google Workspace add-ons. After formulating the hypotheses, a clear user profile will be created, selecting participants who meet the criteria for the defined usability scenario. Participants will be assigned a series of tasks involving Google Workspace add-ons, along with pre-test, in-test, and post-test questions in various experimental studies. Once the data is collected, the hypotheses will be tested using the gathered information, leading to conclusions and necessary insights.

**2.4**. Timescales

Start Date (DD/MM/YY): 07/11/2024

Estimated duration of work: 4 weeks

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*Figure 45 Research Ethics Part A - Application - Part - 2*

**Section 3 - RISK OF HARM**

**NOTE 1: Where indicated below applicants should check if the research will require ethical approval from a National Research Ethics Committee via the Integrated Research Application System (IRAS) - nres.queries@nhs.net.- http://www.hra-decisiontools.org.uk/ethics/**
**NOTE 2: The University of Westminster holds a Human Tissue Authority Licence – This licence is specifically for tissue stored at 115 New Cavendish Street in accordance with the terms of the licence – Advice must be obtained from the University Human Tissue Designated Individual ( N.Presneau@westminster.ac.uk )**

| | RISK OF HARM (to self, colleagues, participants, environment or animals) | Yes | No | N/A |
|---|---|---|---|---|
| 1 | Will any pain or more than mild discomfort result from the study? | ☐ | ☒ | ☐ |
| 2 | Could the study induce any psychological stress or anxiety or cause harm or negative consequences beyond the risks encountered in normal life? | ☐ | ☒ | ☐ |
| 3 | Will the study involve prolonged or repetitive physical or psychological testing of human participants that may put someone at risk, e.g. use of treadmill? | ☐ | ☒ | ☐ |
| 4 | Will the study involve raising sensitive topics (e.g. sexual activity, drug use, revelation of medical history, bereavement, illegal activities, etc.)? | ☐ | ☒ | ☐ |
| 5 | Does your work involve any "relevant material" containing human cells (e.g. blood, urine, saliva, body tissues but NOT established cell-lines) from living or deceased persons (Such work must take account of the Human Tissue Act)? – See Note 1 and 2 above. | ☐ | ☒ | ☐ |
| 6 | Will DNA samples be taken from human participants (Such work must take account of the Human Tissue Act)? – See Note 1 and 2 above. | ☐ | ☒ | ☐ |
| 7 | Does your study raise any issues of personal safety for you or other researchers or participants involved in the project (Especially relevant if taking place outside working hours or off University premises)? | ☐ | ☒ | ☐ |
| 8 | Does your study involve deliberately misleading the participants (e.g. deception, covert observation)? | ☐ | ☒ | ☐ |
| 9 | Does your work involve administration of a food or non-food substance of a different type from or in abnormally higher or lower amounts than normal or one that is known to cause allergic reaction(s) or potential psychological stress? | ☐ | ☒ | ☐ |
| 10 | Does your study involve issues relating to personal and/or sensitive data? | ☐ | ☒ | ☐ |
| | **PARTICIPANTS (and/or their records/associated data) Does your work involve any of the following:** | **Yes** | **No** | **N/A** |
| 11 | Human participants in a health and/or social care setting (e.g. patients, those attending day centres, community care, rehabilitation centres, etc., including in the NHS, other public, private and/or voluntary sectors)? – See Note 1 above. | ☐ | ☒ | ☐ |
| 12 | Human participants who may be deemed vulnerable (e.g. children, people in poverty and/or with physiological or psychological impairments, persons attending rehabilitation centres, persons in easily identifiable positions that could be subject to victimisation, etc.)? | ☐ | ☒ | ☐ |
| 13 | Expectant or new mothers? | ☐ | ☒ | ☐ |
| 14 | Refugees/Asylum seekers? | ☐ | ☒ | ☐ |
| 15 | Minors (under the age of 18 years old)? | ☐ | ☒ | ☐ |
| 16 | Participants in custody (e.g. prisoners or arrestees)? – See Note 1 above. | ☐ | ☒ | ☐ |
| 17 | Participants with impaired mental capacity (e.g. severe mental illness, brain damage, sectioned under Mental Health Act, lowered or reduced sense of consciousness)? – See Note 1 above. | ☐ | ☒ | ☐ |
| 18 | Animals (or animal tissue). | ☐ | ☒ | ☐ |
| | **INFORMATION TO PARTICIPANTS** | **Yes** | **No** | **N/A** |
| 19 | Will you provide participants with a Participant Information Sheet prior to obtaining informed consent which can be taken away by the participant? | ☒ | ☐ | ☐ |
| 20 | Will you describe the procedures to participants in advance, so that they are informed about what to expect? | ☒ | ☐ | ☐ |
| 21 | Will you obtain informed consent for participation (normally written)? OR in the case of using personal data previously acquired was consent given for the reuse of the data for other research purposes? | ☒ | ☐ | ☐ |
| 22 | Will you tell participants that they may withdraw from the research at any time and for any reason without any impact on their care, service provision etc.? | ☒ | ☐ | ☐ |
| 23 | Will you give participants the option of omitting questions they do not want to answer? | ☒ | ☐ | ☐ |
| 24 | Will you tell participants that their data will be treated as confidential and that, if published, it will not be identifiable as theirs? | ☒ | ☐ | ☐ |
| 25 | Will you offer feedback to participants at the end of their participation, upon request (e.g. give them a brief explanation of the study and its outcomes)? | ☒ | ☐ | ☐ |
| 26 | Has external funding or collaboration been applied for/received, which requires | ☐ | ☒ | ☐ |

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*Figure 46 Research Ethics Part A - Application - Part - 3*

| | | | | |
|---|---|---|---|---|
| | institutional ethical consideration or approval? | | | |

**Useful links:**

- http://www.screc.org.uk/ - Social Care Research Ethics Committee
- http://www.hra-decisiontools.org.uk/ethics/ - Human Research Authority decision tool to identify if research needs National Research Ethics Committee approval
- http://www.nres.nhs.uk/applications/guidance/governance-and-directives/?entryid62=131341 – Governance Arrangements for Research Ethics Committees
- http://www.nres.nhs.uk/EasySiteWeb/GatewayLink.aspx?alId=134016 - NRES algorithm "Does my project require review by a Research Ethics Committee"?
- http://www.hta.gov.uk/policiesandcodesofpractice/codesofpractice.cfm - Human Tissue Authority Code of Practice
- http://www.hta.gov.uk – Human Tissue Authority website
- http://www.rsclearn.mrc.ac.uk/MRC_HumanTissueAct/player.html - Medical Research Council online training course for Human Tissue Act.

**What to do next:**

- **If you have answered NO to questions 1-18 (inclusive) and YES to questions 19-25 (inclusive)**, you do not need to complete the Full Research Ethics Approval Form (Part B). Please keep this form for your records, and do not submit to **Faculty Research Ethics Committee** (FREC) unless you require ethical consideration of your study, regardless of ethical implications, by an external body (question 26 has been answered YES). **A list of Faculty contacts is below.**
- **If you have answered YES to any of the questions 1-18 (inclusive) or NO to any of the questions 19-25** the Full Research Ethics Approval Form (Part B) MUST be submitted including Cover Sheet, Part A and Part B of the application form plus any required supplementary documents to the Secretary of the relevant Faculty Research Ethics Committee (FREC). **A list of Faculty contacts is below.**
- If you are applying for external Ethical Approval, please send a *copy* of the Conditions/Approvals letters to the **University Research Ethics Committee** (UREC) Secretary (this may include the original ethical application(s)). Where the external ethics committee/body has equal standing or primary jurisdiction, e.g. another University Research Ethics Committee or a National Research Ethics Committee, any approval will normally be received and noted by the University of Westminster Research Ethics Committee and further consideration may not be required. Where the external committee does not have equal or higher standing than the University Committee then the full ethical approval process at the university may still be required. Additional institutional compliance issues may need consideration by UREC.
- All Applications (dated, signed and authorised) and supplementary information or External Approvals should be sent to the University Research Ethics Committee (UREC) Secretary in *electronic format with a version number, document name and date and the Principal Investigator (or Undergraduate/Postgraduate Taught Student) name. On receipt your application will be issued a unique reference number*
- All new Applications should be submitted to a Research Ethics Committee (FREC or UREC) Secretary a minimum of 10 working days in advance of the Committee meeting date (earlier submission is recommended so that applications can be pre-vetted and obvious issues addressed before the application is considered by the Committee).

**Contact details:**

| Faculty | Chair | Secretary |
|---|---|---|
| Architecture and the Built Environment | Professor Nick Bailey | Colette Davis |
| Media Arts and Design | Dr Anthony Mcnicholas | Fauzia Ahmad |

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*Figure 47 Research Ethics Part A - Application - Part - 4*

| Faculty | Chair | Secretary |
|---|---|---|
| Science and Technology | Dr John Colwell | Mandy Walton |
| Science and Technology Psychology Department Sub Committee | Dr Laura Boubert | TBC |
| Social Sciences and Humanity | Professor Marco Roscini | Victoria Grey-Edwards |
| Westminster Business School | Petar Sudar | Haydn Worley |
| University Research Ethics Committee | Professor Graham Megson | Huzma Kelly |

**For Use in Academic Year: 2015/16**

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*Figure 48 Research Ethics Part A - Application - Part - 5*